

Cyber-Secure SDN: A CNN-Based Approach for Efficient Detection and Mitigation of DDoS attacks

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ABSTRACT

Software Defined Networking (SDN) has become popular due to its flexibility and agility in network management, enabling rapid adaptation to changing business requirements, enhancing network performance, and reducing operational costs. However, the ubiquity of internet-based services has given rise to an alarming increase in cyber-attacks, posing serious threats to the security and stability of modern networks. Among these attacks, Distributed Denial of Service (DDoS) attacks have emerged as one of the most devastating, capable of disrupting critical services. Recent studies have shown that Deep Learning (DL) techniques with Software-defined networking have the potential to mitigate these threats effectively. However, existing solutions suffer from issues such as reliance on pre-defined rules and signatures, computational efficiency, low detection rates, and inefficient notification mechanisms, making them ineffective in detecting DDoS attacks. This paper proposes an efficient approach (BRS + CNN) using Balanced Random Sampling (BRS) and Convolutional Neural Networks (CNNs) to detect DDoS attacks in SDN environments. We have applied various mitigation techniques to mitigate these threats, such as filtering, rate limiting, and iptables rule for blocking spoofed IPs. In addition, we introduce a monitoring system that utilizes rate-limiting to oversee blocked IP addresses, ensuring that legitimate traffic is processed efficiently. The proposed model achieves high performance in binary and multi-classification, with an accuracy of over 99.99% for binary classification and 98.64% for multi-classification. Our proposed DDoS detection system not only detects the attack but also sends detailed contextual information to a designated email address. We compare our model with existing literature and demonstrate its superiority using Area Under The Curve (AUC) analysis. Moreover, we evaluated the efficiency and effectiveness of our proposed DDoS mitigation system by conducting a series of experiments across three distinct scenarios: Attack-Free, Attack-No Mitigation, and Attack-Mitigation. These results demonstrate the robustness of our proposed mitigation system in effectively combating DDoS attacks while also safeguarding the seamless continuity of regular network operations.

1. Introduction

Cybersecurity has become a crucial aspect of modern society. The increasing prevalence of digital communication, cloud computing, mobile devices, and the Internet of Things (IoT) has led to a significant increase in potential attack vectors. Cybersecurity involves safeguarding computer systems and networks from unauthorized access, theft, damage, or disruption of services.

Cyber threats have become more complex and sophisticated, making the need for cybersecurity more critical than ever. Cyber attacks can lead to substantial economic ramifications, damage to reputation, and even the loss of sensitive data. Such attacks can target individuals, organizations, or governments, and their consequences can be far-reaching.

DDoS attacks have emerged as one of the most common cyber attacks and pose a significant threat to online services and businesses. These attacks involve overwhelming a network or server with a massive volume of traffic, making it inaccessible to legitimate users. DDoS attacks can be launched from a botnet, a network of compromised computers, making it challenging to identify the attacker's location (Singh and Behal, 2020). The consequences of DDoS attacks can cause significant financial losses, damage to reputation, and disruptions to online services. They can target a wide range of organizations, from small businesses to large enterprises (Habib and Khursheed, 2022; Najar and Naik, 2022). However, traditional methods of detecting DDoS attacks have proven ineffective in preventing these attacks from overwhelming network resources, and new solutions are needed to mitigate the risk.

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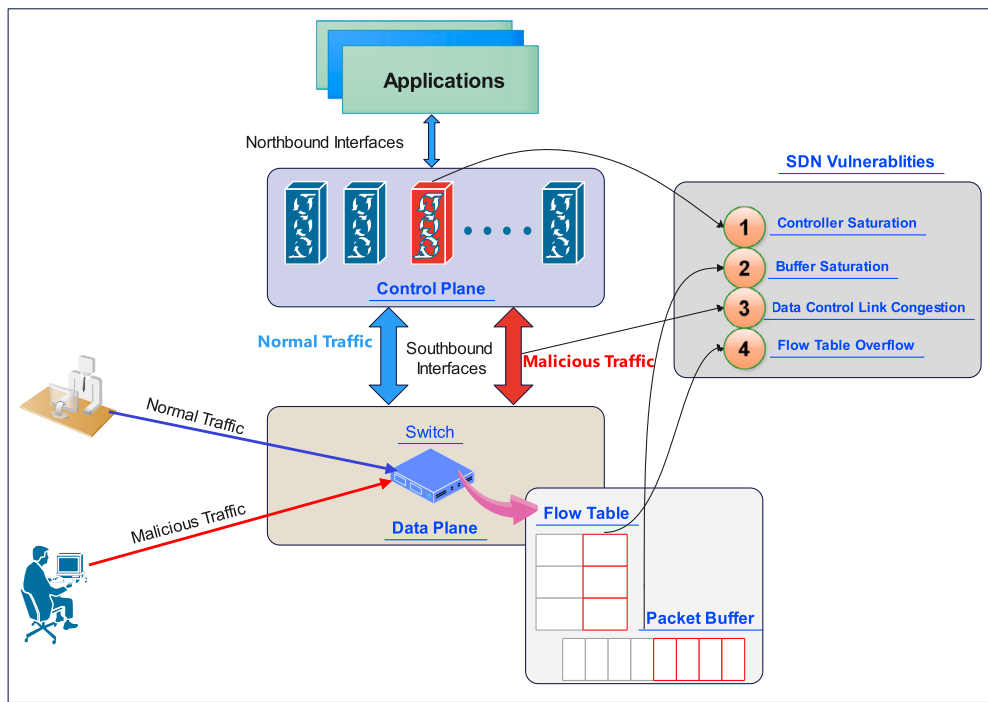


Fig. 1. Conceptual level mapping of vulnerabilities in SDN architecture.

SDN has emerged as a promising technology for improving network security, providing a centralized view of network traffic, and enabling dynamic control of network resources. SDN separates the control plane and data plane in network devices, allowing for centralized management and control of the network. This not only enhances the efficiency of network operations but also provides a more agile and responsive approach to security. The centralized control enables rapid adaptation to emerging threats, making SDN an advantageous and forward-looking solution for bolstering network security in an ever-evolving technological landscape. However, like any technology, SDN is not without vulnerabilities, particularly when it comes to DDoS attacks. The three layers of SDN (application, control, and data plane) are vulnerable to such attacks, as illustrated in Fig. 1. These vulnerabilities include buffer saturation, flow table overflow, data control link congestion, and controller saturation. The article Singh and Behal (2020) discusses in detail the vulnerabilities of the three layers of SDN.

Deep learning, a branch of artificial intelligence, has also shown promise in detecting DDoS attacks by analyzing patterns in network traffic. The success of deep learning models depends on the feature extraction layers used and the size of the training set. According to Hussain and Hnamte (2021), DL technology can comprehend patterns in network behavior, providing insights for attack detection. As networks become more complex, deep neural networks can efficiently analyze network traffic, improving the accuracy of attack detection (Hatcher and Yu, 2018). The convergence of SDN and DL shows the potential to bring about a transformative change in the field of DDoS detection, underscoring its increasing acceptance and exploration in recent academic research pursuits. SDN provides enhanced visibility and control over network traffic, making it an ideal platform for detecting and mitigating DDoS attacks.

To mitigate the risks associated with DDoS attacks, it is essential to have effective detection and mitigation strategies in place, such as intrusion detection systems, firewalls, and traffic filtering (Hnamte and Hussain, 2023). This paper proposes a novel BRS + CNN-based approach that harnesses the benefits of SDN and deep learning to improve the efficiency and accuracy of DDoS attack detection and mitigation in SDN environments. The proposed approach has the potential to provide a more effective and efficient solution for addressing the growing threat

of DDoS attacks in SDN. Through our experiments, we have shown that our proposed approach outperforms traditional DDoS detection techniques in terms of accuracy, computational cost, detection rate, and false positive rate. These results highlight the potential of deep learning in enhancing network security in SDN environments and suggest that our approach can be further developed and deployed in real-world networks to provide better protection against DDoS attacks.

We categorize the main contributions of our work as:

1. A novel BRS + CNN-based efficient DDoS Detection system is proposed with better efficiency, low computational cost, and higher detection rate.
2. The paper also proposes a mitigation system for DDoS attacks, which includes filtering, rate limiting, and iptables rules for blocking spoofed IPs. The mitigation system is evaluated in three different scenarios: Attack-Free, Attack-No Mitigation, and Attack-Mitigation. The results show that the mitigation system is effective in combating DDoS attacks.
3. The paper also introduces a monitoring system that utilizes rate-limiting to oversee blocked IP addresses, ensuring that legitimate traffic is processed efficiently.
4. The proposed system has an inbuilt email notification system. The notification with detailed contextual information is sent to a designated email address.
5. Our model achieved high performance in binary and multi-classification tasks, with a recall, precision, accuracy, and F1 score of over 99.99% for binary classification and a 98.64% performance for multi-classification.
6. We compared the performance of various models with state-of-the-art studies and AUC analysis to demonstrate the superiority of our proposed approach.

The rest of this paper is organized as follows: Section 2 describes the related literature, Section 3 describes the methodology and implementation of the proposed model, Section 4 reports the experimental analysis and results, 5 describes the simulation environment, Section 6 compares the results with prior work, and Section 7 summarily concludes the paper and discusses future work.

2. Related works

The realm of network security has seen extensive research focused on ensuring the safety of existing networks. However, traditional network security measures often lack the required flexibility and scalability to combat the increasing size and frequency of DDoS attacks effectively (Ravi and Shalinie, 2020). In contrast, modern networks such as the OpenFlow-enabled SDN infrastructure offer enhanced flexibility and scalability, enabling security solutions to be tailored to specific network requirements. While the centralized control and programmability of SDN can facilitate faster and more efficient traffic management to mitigate DDoS attacks, the centralized architecture of SDN also renders it more vulnerable to security flaws and targeted attacks than other computing architectures.

To address these challenges, researchers have proposed various security measures, including deep learning-based approaches utilizing Convolutional neural networks for DDoS attack detection in SDN environments. Deep learning has gained significant attention due to their ability to learn hierarchical representations from raw input data, making them well-suited for addressing complex data classification problems such as DDoS attack detection (Yan et al., 2016; Karan et al., 2018; Ali et al., 2020). DL-based approaches reduce the complexity of feature selection and eliminate the need for manual feature engineering, resulting in automated and efficient solutions. Studies have demonstrated that DL-based approaches, particularly those utilizing CNNs, achieve

high detection accuracy in identifying DDoS attacks within SDN environments while maintaining low computational costs, complexity, and latency. Recent research efforts have specifically focused on utilizing DL approaches, especially CNN, for identifying DDoS attacks in SDN environments, highlighting their potential as a promising solution for DDoS attack detection.

For our literature review, we conducted a systematic search of papers published between 2019 and 2023 in top-level databases such as Science Direct, Google Scholar, and IEEE Xplore. After screening and analyzing a large number of papers, we selected 25 high-quality papers that best align with our research objectives. These papers will provide valuable insights and an in-depth understanding of the topic under investigation. In addition, we implemented and tested five papers to verify their results and performance. The analysis of these papers not only provided valuable insights into optimizing the system's performance but also helped us to develop a more reliable and efficient solution for mitigating DDoS attacks in SDN environments.

Table 1 summarizes a comprehensive literature review of DDoS detection methods proposed and utilized in the past (Elmasry et al., 2020; Pérez-Díaz et al., 2020; Erhan and Anarim, 2020; Ujjan et al., 2020; de Assis et al., 2020; Novaes et al., 2020a; Wei et al., 2021; Elsayed et al., 2020a; Wang and Li, 2021; Choobdar et al., 2021; Ferrag et al., 2021; Zainudin et al., 2023; Amaizu et al., 2021; Musumeci et al., 2022; Maheshwari et al., 2022; Mishra et al., 2023; Zhou et al., 2023; Chouhan et al., 2023; Liu et al., 2023). The analysis presented in

Table 1

Existing literature summary: overview of machine learning, deep learning, and hybrid approaches.

Author and Year	Dataset Used	Methodology	Main Findings	Constraints
1. Elmasry et al. (2020)	CICDDoS2017 and NSL-KDD	Double Particle Swarm Optimization (PSO) metaheuristic for evolving deep learning architectures	Experimental results show a significant improvement in detection rate and reduction in false alarm rate by 4-6% and 1-5%, respectively.	The use of complex Deep Belief Networks (DBNs) with more hidden layers can result in slower training and inference times.
2. Pérez-Díaz et al. (2020)	CICDDoS2017	Multilayer Perceptron (MLP)	Proposed architecture for LR-DDoS attack detection and mitigation in SDN environments. Achieves a 95% detection rate for LR-DDoS attacks. Architecture deployed in a real virtualized environment using ONOS controller and Mininet.	It is specifically designed for detecting low-rate DoS attacks; still, due to the inherent complexity of detecting such attacks, its detection accuracy of 95% may not be considered particularly high.
3. Erhan and Anarim (2020)	TCP and UDP flood datasets	Matching pursuit and wavelet techniques	Achieves detection efficiency (95.3–95.7%) with True Positive Rate (TPR) (80–81%) using multiple characteristics of network traffic.	Suboptimal approximation, hybrid detection framework increases complexity, and Efficient feature selection is needed to enhance model performance and reduce computational complexity
4. Ujjan et al. (2020)	Generated Dataset	Combines sFlow and adaptive polling-based sampling with Snort IDS and deep learning model.	Achieves detection accuracy (95%), low False Positive Rate (FPR) (less than 4%), Efficiently mitigates various DDoS attacks in IoT network.	High computational complexity, centralized deployment makes it more vulnerable to attacks (single-point failure).
5. de Assis et al. (2020)	CICDDoS2019	CNN	Proposed a real-time SDN security system using CNN and game-theoretical approaches for DDoS detection and attack mitigation. Achieved an accuracy of 95.4% with low FPR.	Limited number of DDoS attack types, Requires a significant amount of training data to detect and mitigate attacks accurately; Further validation in real-world scenarios is needed, Lack of a notification mechanism.
6. Novaes et al. (2020a)	CICDDoS2019	Long Short-Term Memory (LSTM-FUZZY)	DDoS and Portscan attacks detection using a Fuzzy inference system, Analysis of the network traffic performed in near realtime, Achieved precision of 97.89%.	Limited to two classes only, no alert and monitoring system, performance evaluation is limited to a few metrics.
7. Wei et al. (2021)	CICDDoS2019	Autoencoder (AE-MLP)	Hybrid deep learning model, Achieved an accuracy of 98.34% and F1-score exceeding 98%.	High response time, limited to 5 classes only, Efficient data preprocessing techniques need to be explored, lack of a simulation environment, The proposed model focuses exclusively on attack detection and doesn't address mitigation strategies.
8. Elsayed et al. (2020a)	InSDN	LSTM-based AE	Outperforms standalone One-Class Support Vector Machine (OC-SVM) method with 90.5% accuracy and 93% precision, recall, and F1 scores.	Long training time (147.58 sec), a limited number of classes in the dataset, and use of irrelevant features increase computational cost without improving accuracy.

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Table 1 (continued)

Author and Year	Dataset Used	Methodology	Main Findings	Constraints
9. Wang and Li (2021)	CICDDoS2019	Transformer-based DDoS TC model	The hybrid model demonstrates exceptional performance, achieving accuracy levels of 0.9982% for an 8:2 training/test ratio, 0.9979% for a 7:3 ratio, and 0.9974% for a 6:4 ratio.	Does not discuss the computational resources required, Evaluation of the model's performance is based on a random division of the training set and test set, suboptimal recall rate when the train/test ratio was 6:4, High computational cost.
10. Choobdar et al. (2021)	CICIDS2017	Deep Neural Networks (DNN)	Stacked AEs for feature extraction and classification, Comparative analysis using realistic datasets (NSL-KDD, CICIDS2017) to validate results, and addressing multi-classification, Achieved an accuracy of 98.5% and 94.57%.	Requires pre-training using sparse stacked AEs, which may require a significant amount of training data and computation resources, Focuses on detection only, No alert notification system.
11. Ferrag et al. (2021)	CICDDoS2019	Recurrent Neural Network (RNN), CNN and DNN	Evaluating the performance of three deep learning models for distinguishing DDoS attack traffic from benign traffic in the context of Agriculture 4.0, Achieved an accuracy of 95.12%, 94.88%, and 93.88%.	Limited availability and quality of datasets for training and testing the proposed IDS models. The proposed model focuses exclusively on attack detection and doesn't address mitigation strategies.
12. Zainudin et al. (2023)	CICDDoS2019	Hybrid DNN	Low-cost approach using Extreme Gradient Boosting (XGBoost) feature selection and a hybrid CNN-LSTM model for classification, Achieved an accuracy of 99.50%.	Resource constraints on Industrial Internet of Things (IIoT) devices, Complexity and interoperability, limited to binary classification, The model only detects attacks and doesn't handle mitigation.
13. Amaizu et al. (2021)	CICDDoS2019	Hybrid DNN Models with Pearson Correlation Coefficient (PCC) Feature Extraction for 5G/B5G DDoS Detection	The proposed framework achieved an accuracy rate of 99.66% and a loss of 0.011. All models, except for a CNN ensemble, were surpassed by the performance of the suggested approach.	The proposed framework primarily focuses on detection without addressing mitigation strategies, The complexity of the model could lead to longer detection times, This could impact the framework's performance in real-time environments.
14. Musumeci et al. (2022)	Generated Dataset	Machine Learning-DAD	Explored integration of P4 programmable technology with machine learning techniques for real-time detection systems, Proposed DDoS Attack Detection (DAD) approach to identify SYN flood attacks on SDN switches with a 98% overall performance.	Focuses on local detection of Synchronize (SYN) DDoS flood attacks within the data plane, and It does not cover broader implications for DDoS attacks on SDN controllers. The model is trained and tested on a compact dataset, Lacks a built-in mitigation system.
15. Maheshwari et al. (2022)	CICDDoS2019	Hybrid Meta-Heuristic Optimization	Proposed novel Hybrid Meta-Heuristic Optimization (BHO) for optimal feature selection, An ensemble method with six fundamental classifiers was used, A testing platform was constructed using Mininet and POX controller, Achieved an accuracy of 99.41% on the CIC-DDoS and 99.36% on the CAIDA datasets.	The limitations of the proposed model include a lack of clear details about the mitigation strategy, the absence of a notification system for alerting users/administrators upon attack detection, and an insufficient discussion on the management of blocked Internet Protocol addresses (IPs) to prevent unintended disruption of legitimate traffic.
16. Mishra et al. (2023)	CICDDoS2019	Extra Tree Classifier	Proposed framework for DDoS attack categorization and prediction. Achieved 99.87% accuracy using AdaBoost Classifier, Utilized Extra Tree Classifier for feature optimization, resulting in 25 key attributes.	Limited classification to 8 out of 13 classes, Mitigation strategies not extensively discussed.
17. Zhou et al. (2023)	Synthetic	Decision Tree (DT), DL, K-Nearest Neighbors (KNN), Random Forest (RF), and Support Vector Machine (SVM)	Introduces five new features extracted from heterogeneous packets, Demonstrates consistent effectiveness of proposed features across various DDoS attack types, Shows substantial accuracy improvements (21% to 53%) over existing methods.	The proposed model primarily focuses on detection without addressing mitigation strategies, It is a signature-based method, so it cannot detect new or unknown DDoS attacks, tested against a limited number of DDoS attacks.
18. Chouhan et al. (2023)	Generated Dataset	SVM, RF, K-NN, Extreme Gradient Boosting, Naive Bayes (NB)	SVM outperformed other classifiers with 99.398% accuracy, 99.413% precision, 99.397% recall, 0.718% FAR, 0.995 AUC, and 99.400% F1 value. The deployment of SVM in the SDN controller allowed for real-time detection of attacks in the ongoing traffic.	The real-time performance of the proposed model in real-world environments could be impacted by longer detection times due to the complexity of the model structure. The dataset used in this paper is relatively small, Additionally, the model primarily focuses on detection and does not address mitigation.
19. Liu et al. (2023)	CSE-CIC-IDS2018	Feature extraction, Binary Grey Wolf Optimization, Supervised learning (SVM, RF, DT, XGBoost, KNN)	Random forest performs best across performance metrics and achieves 99.13% of accuracy for binary classification, The proposed model mitigates DDoS attacks by applying traffic modification rules.	The proposed method may not be able to detect all types of DDoS attacks, Relies on packet dropping for mitigation without clear instructions on handling dropped packets, Lacks effective strategies for detecting and mitigating spoofed packets, and Lacks an alert notification system.

the table includes dataset description, methodology, main findings, and the constraints associated with these methods, which offers valuable insights into the existing approaches for combating DDoS attacks.

2.1. Research gap

After reviewing the available literature, it is evident that several traffic features, attributes, and variations have been utilized to distinguish normal and attack traffic. While traditional DDoS intrusion detection systems using machine learning approaches suffer from issues such as reliance on pre-defined rules and signatures, low accuracy results (Pérez-Díaz et al., 2020; Hezam et al., 2022; Aamir and Zaidi, 2021; Alghazzawi et al., 2021), and expensive and complex hardware infrastructure (Gopi et al., 2021; Alghazzawi et al., 2021).

Although most research has focused on detecting DDoS attacks using various methods, there is a critical gap in the subsequent mitigation of these attacks once they are detected. Most existing studies do not address the issue of what to do with attack flows, which can also impact legitimate communications within the network and require additional resources, especially for small-scale networks (Dayal and Srivastava, 2023). Additionally, there is a notable gap regarding the efficiency and scalability of monitoring mechanisms used in mitigation algorithms. Therefore, it is essential to develop mitigation approaches tailored to specific network requirements.

In contrast, deep learning methods have demonstrated superior performance to existing machine learning approaches in DDoS detection. These techniques can automatically extract relevant features from the data, resulting in more accurate detection and reduced false positives. However, deep learning methods also have some challenges, such as computational complexity due to the intricate architecture of hybrid models, processing overhead (Dehkordi et al., 2020; Alghazzawi et al., 2021), outdated and invalid datasets (Ahuja et al., 2021), inability to detect all DDoS attacks (Eswari, 2021; Awan et al., 2021), long execution time (Yuan et al., 2017; Pelletier et al., 2019), and the lack of an inbuilt notification system. Therefore, there is a need for the development of more efficient and streamlined DDoS detection systems that address these issues.

3. Methodology and implementation

In a world where cyber threats are constantly evolving and posing a serious risk to businesses and organizations, it is crucial to develop robust methods for detecting and mitigating these threats. In this study, we propose an innovative BRS + CNN based approach for the detection and mitigation of DDoS attacks in SDN networks. Our methodology comprises four modules (Fig. 2):

- Module 1: Data Preprocessing and Model Construction
- Module 2: DDoS Detection System
- Module 3: DDoS Mitigation and Monitoring System
- Module 4: Email Notification System

In order to detect DDoS attacks in a switch, a Deep learning algorithm (CNN) is employed by the controller. If a DDoS attack is identified, the switch enforces rules to block the attacker and remove their flow entry. However, this may cause legitimate hosts to have their benign packets discarded if they attempt to transmit data to a host whose address has been spoofed by the attacker and subsequently blocked. To address this, a dedicated monitoring system was implemented to oversee the blocked IP addresses and determine if the attacker has ceased using them, allowing genuine packet requests to be processed. This enhances the reliability and availability of SDN.

To further fortify network security and counter IP spoofing attempts and DDoS attacks, an IP Spoofing Detection and Mitigation mechanism is implemented. The Algorithm 3 checks incoming packet source IPs

against a list of valid addresses. Unauthorized IPs are identified as potential spoofing attempts and blocked using iptables, a powerful Linux firewall tool. To optimize IP blocking without disrupting legitimate traffic, a rate limit mechanism is employed. Algorithm 3 continuously monitors and updates the list of blocked IP addresses, providing real-time insights to administrators regarding blocked IPs and potential security threats. This additional module enhances the overall security and operational efficiency of the SDN environment. In addition, when a DDoS attack is detected, a detailed notification email is sent to the administrator's registered email address. This information will help administrators to know more about the attack and can enhance the system for future challenges.

3.1. Data preprocessing and model construction

Data pre-processing is a crucial step in deep learning that can significantly impact the model's performance. It involves cleaning, normalizing, selecting relevant features, and augmenting the dataset to improve its quality and suitability for the model. The primary objective is to prepare the data by removing noise and inconsistencies, enhancing the model's ability to learn and make accurate predictions. We addressed noise, removed null features, and eliminated duplicate rows in the CICDDoS2019 dataset.¹ Additionally, we implemented measures to reduce the size and balance the dataset, ensuring a more robust and representative training environment for our model. We created a balanced dataset using random sampling by taking equal records from each file and concatenating pre-processed files. We also implemented pre-processing and feature selection techniques like removing duplicates, constants, and features with low information gain (threshold ≤ 0.05). This led to the extraction of 58 features, and we applied min-max normalization and partitioned the dataset into training and testing sets. The training set consists of more than 4.9 lakh records, the validation set consists of more than six thousand records, and the testing set consists of more than 1.2 lakh records.

The following outlines the respective preprocessing operations conducted on the dataset:

- To start with feature elimination, it was identified that the original dataset consisted of more than 50 million records with 88 attributes and encompassed 13 attack types. Recognizing similarities between the DNS and LDAP classes, the LDAP class was excluded from the evaluations. Following feature elimination, it became apparent that certain attributes in the dataset solely contained zero or null values. These attributes were consequently removed, leading to the creation of a new dataset containing 66 features.
- To address the imbalance issue and reduce the size of the dataset, we employed a balanced random sampling approach. Initially, we observed that the dataset contained an uneven distribution of records between benign and attack classes. Subsequently, we randomly selected an equal number of samples from each attack type, matching the number of benign samples (56,025). This balanced sampling strategy ensures that the model is trained on an equal set of samples from each class, preventing it from being biased towards the majority class. Consequently, the model becomes more adept at recognizing patterns across all classes, enhancing its overall performance and generalization capabilities. The new dataset consists of 6,72,300 records with 66 attributes.
- Our choice of using random sampling was motivated by the dual objectives of reducing the dataset size and achieving class balance. With 56,025 benign samples as the benchmark, we maintained this count for each attack category during the random sampling process. In order to eliminate bias towards the majority class and to promote robust learning across all categories, this balanced

¹ <https://www.unb.ca/cic/datasets/ddos-2019.htm>.

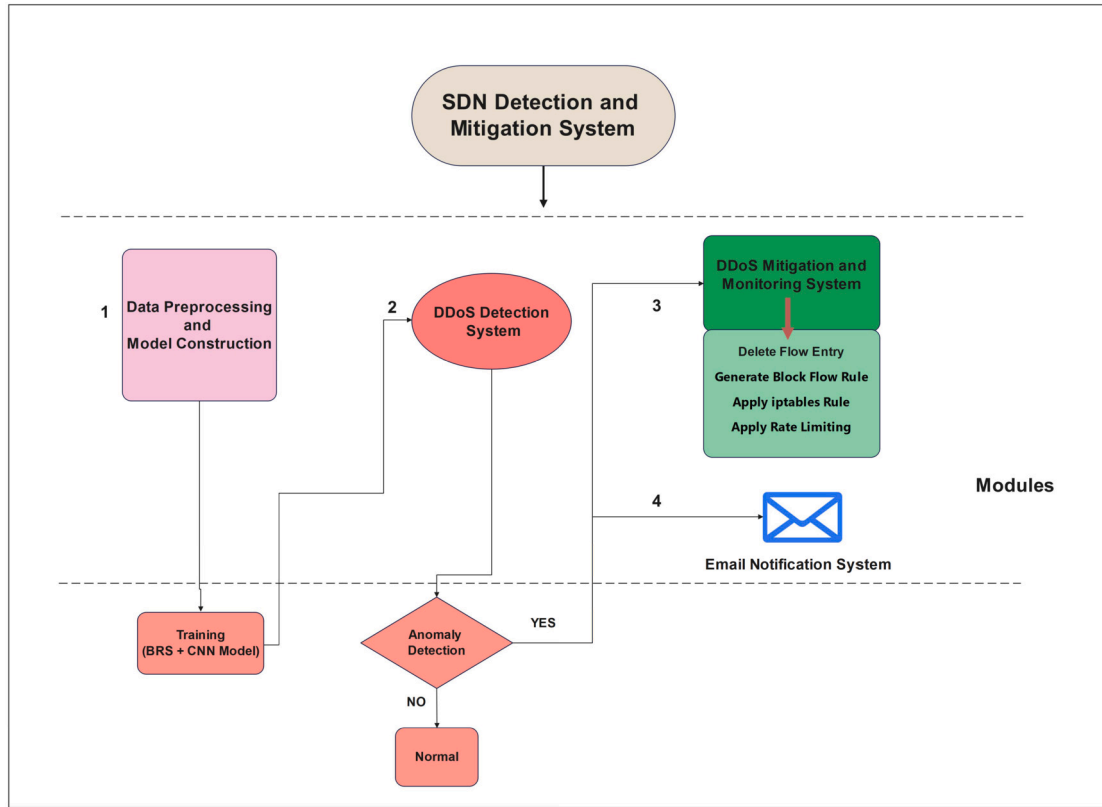


Fig. 2. Modular structure of SDN Detection and Mitigation system.

sampling technique makes sure that the model is trained on a representative collection of samples from each class. The results demonstrated remarkable effectiveness in both binary and multi-classification, with accuracy rates reaching 99.99% and 98.64%, respectively. Moreover, the computational cost remained impressively low. For binary classification, we achieved an inference time of 0.2633 seconds, and for different types of DDoS attacks, we achieved an inference time of 0.3697 seconds.

- Then, we employ the Info Gain feature evaluation algorithm to streamline the dataset, crafting a more compact yet high-performing dataset. Following this process, we arrive at a final set of 58 features, including the label. While various algorithms from the existing literature underwent testing, the Info Gain algorithm emerged as the preferred choice due to its consistently superior performance compared to other selection methods.
- Finally, we apply min-max normalization to the newly crafted balanced dataset. This transformative process yields the final dataset, meticulously prepared for utilization in both training and testing operations.

Through meticulous data preprocessing, we aimed to refine the dataset for enhanced training effectiveness and superior detection and classification performance. To address the large number of records and significant class imbalance, we employed feature elimination, balanced random sampling, and the Info Gain algorithm. This strategic approach resulted in a more compact and balanced dataset, fostering improved model performance and addressing computational efficiency concerns.

The model architecture (Fig. 3) comprises 2 Conv1D layers, a Max-Pooling1D layer, a ReLU activation function, and 2 Dense layers with 128 and 64 units. A dropout rate of 0.5 is used to prevent overfitting, and the output layer has 12 units, using a softmax activation function to produce a probability distribution. We trained the model using the Adam optimizer with a learning rate of 3e-4 and evaluated using accuracy as the metric. Hyperparameter tuning is essential for deep learning,

Table 2

Model construction and hyperparameters.

Model Construction	Details
Number of Conv1D Layers	2
Filter Size	64
Kernel Size	3
MaxPooling1D Layer	Yes
Dropout Rate	0.5
Number of Dense Layers	2
Number of Units in Dense Layers	128, 64
Output Layer Activation Function	Softmax
Loss Function	Categorical cross-entropy
Optimizer	Adam
Learning Rate	3e-4
Hyperparameters	Values
Learning Rate Schedule	Exponential decay
Momentum	0.9
Number of Epochs	50
Batch Size	256
Layer Type	Conv1D
Activation Function	ReLU

and we carefully selected and tuned hyperparameters to achieve the best performance of the CNN model. The details of these parameters are shown in Table 2.

3.2. DDoS detection system

In this proposed work, detecting DDoS attacks promptly and precisely is crucial in the context of Software-Defined Networking. To achieve this goal, an efficient deep-learning approach is utilized. Specifically, Convolutional Neural Networks (CNNs) are employed to select features that are most effective for the DL model. An efficient feature selection technique is employed to choose 58 features along with their labels from a total of 88 features. The selected features are then used to train the CNN model, which enables accurate detection of DDoS attacks in real-time.

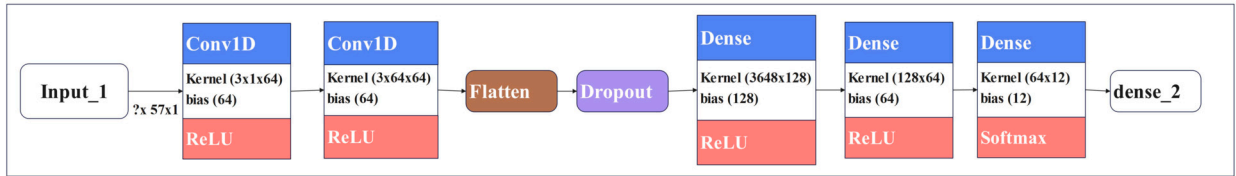


Fig. 3. Proposed model CNN architecture.

Algorithm 1 summarizes the proposed approach. The main aim of this algorithm is to detect DDoS attacks in a network environment by analyzing flow statistics collected from active OpenFlow switches over a time window of length ΔT . The algorithm extracts features from the flow statistics using a feature extractor function. Then it uses a pre-trained classifier model to classify the features as normal or attack traffic. When an attack is detected, the algorithm increments an alert counter and invokes a mitigation function to prevent further damage. The algorithm effectively distinguishes normal and attack traffic, blocking suspicious traffic flows while allowing normal traffic flows to pass through the network.

Algorithm 1 DDoS Detection Algorithm.

Require: Time window ΔT , feature extractor function F , pre-trained classifier model M , mitigation function G
Ensure: Attack or Normal

- 1: **for** each active OpenFlow switch in the network during ΔT **do**
- 2: Send Flow Statistics Request to the switch via the controller
- 3: Collect the Flow Statistics from the switch
- 4: Extract Feature Set f from the Flow Statistics using F
- 5: Classify f using M to obtain the predicted label y
- 6: **if** $y = \text{Attack}$ **then**
- 7: Call the mitigation function G
- 8: **end if**
- 9: Allow the flow to access the host
- 10: **end for**

Overall, this algorithm is designed to provide a flexible and effective framework for detecting and mitigating DDoS attacks in a network environment. By monitoring network traffic in real-time and using deep learning algorithms to classify traffic, the algorithm can quickly identify and respond to DDoS attacks before they cause significant damage to the network.

3.3. DDoS mitigation and monitoring system

DDoS attacks are a major threat to Software-Defined Networks (SDNs). By overwhelming network resources and rendering services inaccessible, DDoS attacks can disrupt network stability and user experience. To safeguard against these disruptive attacks, it is essential to employ robust mitigation techniques within the SDN framework. In this study, we have used a combination of mitigation techniques to combat the diverse forms of DDoS attacks in SDN environments.

Our mitigation strategies aim to ensure the continuity of network services by countering the adverse impacts of malicious traffic influx. Specifically, we have focused on the following key techniques:

3.3.1. DDoS mitigation techniques utilized in proposed approach

- **Filtering:** Filtering is a network security strategy that allows administrators to control the passage of data packets within a network. It can be used to permit or block traffic based on specific criteria, such as source and destination IP addresses, protocol, or ports. This helps to protect sensitive resources and systems by restricting unauthorized or potentially harmful network traffic. For example, SDN controllers can be used to filter traffic based on source IP address, destination IP address, and port number to prevent DDoS attacks.

- **Rate-limiting:** Rate limiting is a network management technique that prevents DoS attacks by limiting the amount of traffic from a single IP address in a given period of time. It works by imposing a predefined threshold on the rate of data transmission. If an IP address exceeds this threshold, it will be blocked from sending further traffic. It can be applied at various levels, such as by IP address, port, or application. This allows network administrators to fine-tune the rate-limiting policy to meet the specific needs of their network.
- **Spoofed IP detection:** Spoofing detection is a security mechanism that helps to prevent DDoS attacks and other malicious activity by checking the source IP address of packets to see if it is valid. If the source IP address is not valid, the packet is dropped. Spoofing detection mechanisms work by analyzing the characteristics of incoming network traffic and looking for inconsistencies or irregularities that indicate a potential spoofing attempt. For example, a spoofing detection mechanism might look for packets with a source IP address not in the same subnet as the destination IP address or a source IP address not used by any known network device.
- **iptables:** iptables is a powerful tool that can be used to control network traffic in Linux systems. It works as a packet-filtering firewall, which defines rules governing the flow of data packets entering, exiting, or traversing the system. These rules can be used to control access, enhance security, enable Network Address Translation (NAT), and forward traffic between different ports. Iptables rules are organized into tables and chains. Tables group related rules together, and chains order the rules in a specific way. The filter table is the most commonly used, which controls incoming and outgoing traffic. The other tables are the nat table (for NAT), the mangle table (for packet modification), and the raw table (for advanced packet filtering).
- **Monitoring:** Monitoring is important to any DDoS mitigation strategy. It can be used to track the effectiveness of mitigation techniques, identify new threats, and ensure that legitimate traffic is not blocked. For example, monitoring can be used to track the number of packets that are being dropped by the rate limiter. This information can be used to determine whether the rate limiter is configured correctly and whether it is effective in mitigating DDoS attacks. It can also be used to track flow entry requests from blocked IP addresses. This allows the controller to see if legitimate traffic is sent from the blocked IP address. If the controller sees legitimate traffic being sent from the blocked IP address, it will not block the IP address. This will ensure that legitimate traffic can still reach the network.

The Mitigation Algorithm protects the network from DDoS attacks by inserting a block flow rule on the switches. The Mitigation Algorithm employs two effective strategies to prevent DDoS attacks. Firstly, it removes the flow entry for the attacked switch, which clears the congestion and prevents the attacking IP from approaching it further. Once the attack is detected, the attack information is passed to the controller.

The controller then blocks the attacking IP if the number of packets sent by the attacker to the switch exceeds the rate limit, which denies all data packet requests from that IP address. For the existing mitigation mechanism, this has become an issue. To address this, the controller uses a monitoring system to track flow entry requests from blocked

IP addresses, ensuring that legitimate packets are still accepted. Once the attack stops, the system will acknowledge and remove the IP address from the block list, enabling genuine packets to be received again. The mitigation mechanism is explained in Algorithm 2 below. The algorithm repeats steps 2-7 for each switch in the network that needs to be configured. Our proposed model's overall detection and mitigation mechanism is outlined in Fig. 4.

Algorithm 2 DDoS Mitigation Algorithm.

Require: *Switch*: the OpenFlow switch
Require: *VictimIP*: IP address of the victim
Require: *RateLimit*: maximum packets per second
Ensure: Block flow rule on switches to prevent further attack traffic
1: **for all** *switch* in the network **do**
2: Identify the switch that needs configuration
3: Delete any existing flow entry related to *VictimIP*
4: **if** number of packets sent by attacker to *switch* > *RateLimit* **then**
5: Block the attacker
6: Generate a block flow rule with *VictimIP* as the destination address
7: Write the flow rule onto *switch* to take effect
8: **end if**
9: **end for**

Overall, the Mitigation Algorithm is an essential component of any DDoS detection and mitigation system. By removing the victim's IP address from the network, the DDoS attack is prevented from causing further harm. Without such mitigation measures, a DDoS attack can lead to severe disruption of network services and potentially result in significant financial losses. Therefore, the mitigation algorithm plays a crucial role in ensuring the availability and integrity of network services, protecting businesses and individuals from the negative impacts of DDoS attacks.

3.3.2. IP spoofing detection and mitigation

The proposed IP spoofing detection and mitigation system presents a robust and efficient approach to fortify network security against IP spoofing attempts and Distributed Denial of Service attacks.

Algorithm 3 for IP spoofing detection and mitigation operates as follows: When an IP packet is received, the algorithm first checks the Source IP address (*src_ip*) of the packet. To determine the legitimacy of the source IP, the algorithm maintains a list of Valid IP addresses (*valid_ip_range*) that are considered authorized and allowed on the network. If the *src_ip* is not present in the valid IP range, the algorithm identifies it as a potential IP spoofing attempt and logs a warning message to indicate the detection of a spoofed packet.

The algorithm proceeds to take swift and proactive action to block the source IP address of the detected spoofed packet. It achieves this by utilizing iptables, a powerful firewall tool in Linux systems. The algorithm runs a subprocess command to add a blocking rule in the INPUT chain of iptables, instructing it to drop any incoming packets from the detected spoofed IP address. By doing so, the algorithm effectively neutralizes the threat posed by the spoofed IP.

To maintain visibility into the detection and blocking process, the algorithm prints the list of spoofed IPs and blocked IPs. This allows administrators to monitor the status of blocked IPs and track any suspicious activity. Furthermore, to optimize the IP blocking process and avoid unnecessary disruptions to legitimate traffic, the algorithm employs a rate limit mechanism to optimize IP blocking and prevent unnecessary disruptions to legitimate traffic. It checks the packet count received from a specific source IP within a defined time window against the rate limit. If the packet count is below the limit, the algorithm considers the traffic legitimate and takes no action to block the IP. This approach ensures false positives are minimized, and legitimate users are not mistakenly blocked. However, when the packet count exceeds the rate limit, indicating a potential DDoS attack, the algorithm initiates IP mitigation. It adds a blocking rule using iptables to drop incoming packets from the detected source IP, effectively neutralizing the threat posed by the spoofed IP.

As part of its continuous monitoring and dynamic updating, the algorithm periodically displays and updates the list of blocked IP addresses (*srcip_blocked*). This real-time information gives administrators precise insights into the status of blocked IPs, allowing them to adapt their security measures to the evolving threat landscape.

In summary, the algorithm's strength lies in its ability to promptly detect IP spoofing attempts and DDoS attacks, along with its proactive approach to blocking suspicious IPs using the robust capabilities of iptables. By employing a rate limit mechanism, the algorithm can accurately identify and mitigate DDoS attacks while ensuring that genuine traffic is not inadvertently blocked. This ensures the highest level of security and operational efficiency, safeguarding the network from potential threats and minimizing disruptions to legitimate traffic.

Algorithm 3 IP Spoofing Detection and Mitigation.

Require: *src_ip*: Source IP address of the packet.
Require: *valid_ip_range*: A set of valid IP addresses.
Require: *srcip_blocked*: Dictionary containing blocked IP addresses.
Require: *rate_limit*: Maximum packets from an IP in a time window.
Require: *web_packet*: Dictionary with packet count for each source IP.
1: **if** *src_ip* not in *valid_ip_range* **then**
2: **Output:** "IP Spoofing detected with IP: *src_ip*".
3: **end if**
4: **if** *src_ip* not in *srcip_blocked* **then**
5: Execute: `os.system("sudo iptables -A INPUT -s %s -j DROP" % src_ip)`.
6: **Output:** "IP *src_ip* has been mitigated using iptables".
7: Add *src_ip* to *srcip_blocked*: *srcip_blocked*[*src_ip*] = "Blocked".
8: **end if**
9: **if** *src_ip* in *srcip_blocked* & *src_ip* in *web_packet* **then**
10: **if** *web_packet*[*src_ip*] < *rate_limit* **then**
11: Remove *src_ip* from *srcip_blocked*.
12: **end if**
13: **end if**
14: **Output:** "Spoofed IPs:", *srcip_blocked*.
15: **Output:** "List of Blocked and Spoofed IPs:", *spoofed_ips_mitigated*.

3.4. Email notification system

We incorporate an alert feature into our IDS. Unlike traditional approaches that may involve expensive third-party internet security providers, we have adopted a cost-effective solution by utilizing a free email service to implement this feature. In contrast to many IDSs that store alerts within log directories, our system takes a proactive approach by instantly sending notifications to a designated Gmail account (see Fig. 5) along with the details of the attack, such as the source IP, blocked and spoofed IPs, bandwidth consumption, and timestamp. This proactive alerting eliminates the need for manual log monitoring, saving valuable time and resources. Additionally, to uphold the confidentiality and integrity of the communication, we have implemented an extra layer of security by encoding the email content using a base64 encoder. This ensures that the sensitive information remains protected during transmission.

4. Experimental analysis and results

Table 3 presents the experimental setup, including system specifications, software used, simulation details, and a hyperlink to the code repository.

4.1. Selection of the dataset

We utilized the CICDDoS2019 dataset in our study (Unb-cic ddos, 2019), which is designed to provide reliable information for training DDoS attack detection models. This dataset addresses some of the shortcomings of previously proposed datasets, such as the lack of categorized information, the ability of attackers to execute simple-level intrusions,

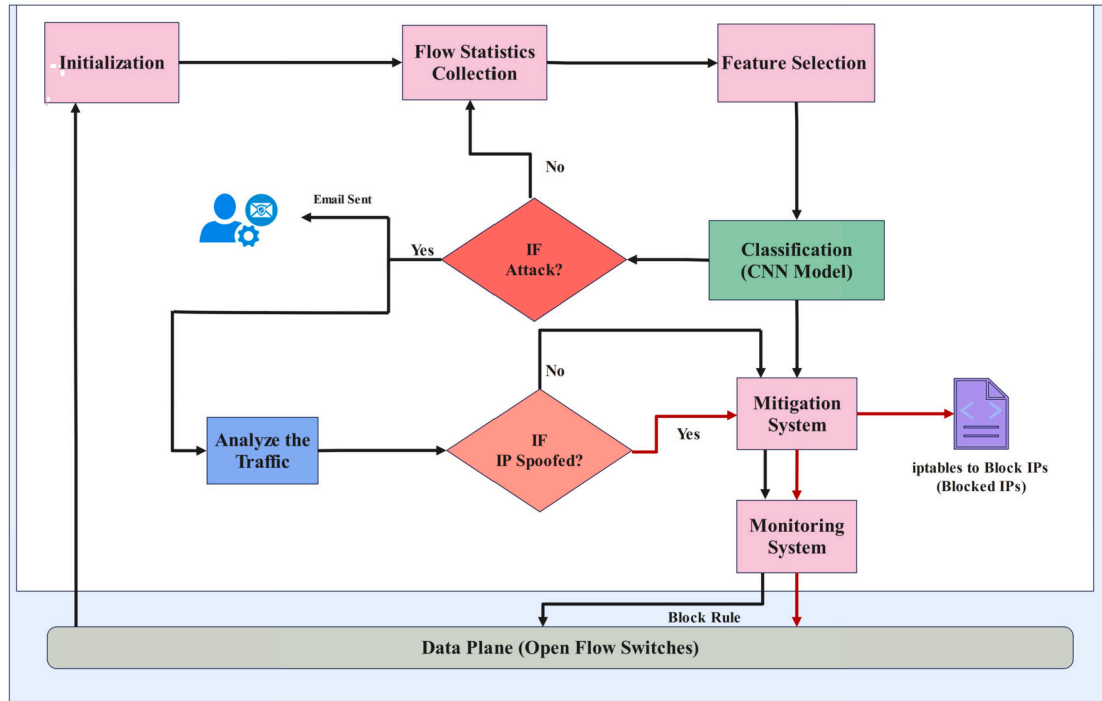


Fig. 4. Proposed DDoS detection and mitigation system.

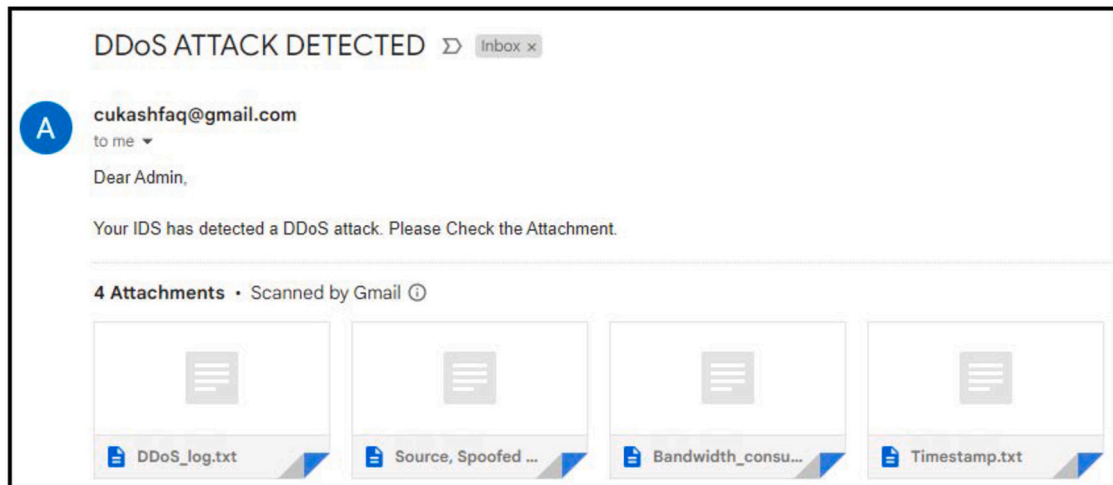


Fig. 5. Email notification system.

Table 3
System specifications and software used for experiments.

System Specifications	Details
Workstation	Windows-based 11 Pro workstation
CPU	Intel(R) Xeon(R) W-2155 running at 3.30 GHz and 3.31 GHz
RAM	32 GB
GPU	NVIDIA Quadro P4000 with 8 GB of GPU RAM
IDE Platform	Pycharm Community
Packet encoding Software	base-64
	Python 3.9.13, Numpy, Keras, Pandas, Sklearn, and Matplotlib, Email, Scapy, Smtplib, email-encoder, Datetime
Simulation	mininet 2.3.0 installed on a Virtual Box supporting OpenFlow version 2.9.8, Pox controller
Code repository	https://github.com/AshfaqNajar/SDN-DDoS-Detection.git

and the inability to generate modern attacks such as MSSQL, NETBIOS, UDP, UDPLAG, and other similar attacks.

The CICDDoS2019 dataset contains 11 CSV files generated by CFlowMeter containing invalid values such as NaN, blank cells, and infinite values, which can negatively impact the model's accuracy. Despite this issue, the dataset is still deemed sufficient for model training and validation, consisting of 5,00,63,112 records and 88 attributes. The dataset includes 56,863 normal traffic samples and 5,00,06,249 attack traffic samples, each with 88 common attributes.

To address the class imbalance problem, we created a balanced dataset using random subset selection. This balanced sampling strategy ensures that the model is trained on an equal set of samples from each class, preventing it from being biased towards the majority class. This approach, in turn, minimizes the risk of overtraining and enhances the model's capacity to generalize across diverse instances, thereby bolstering its overall predictive performance. Additionally, data correlation

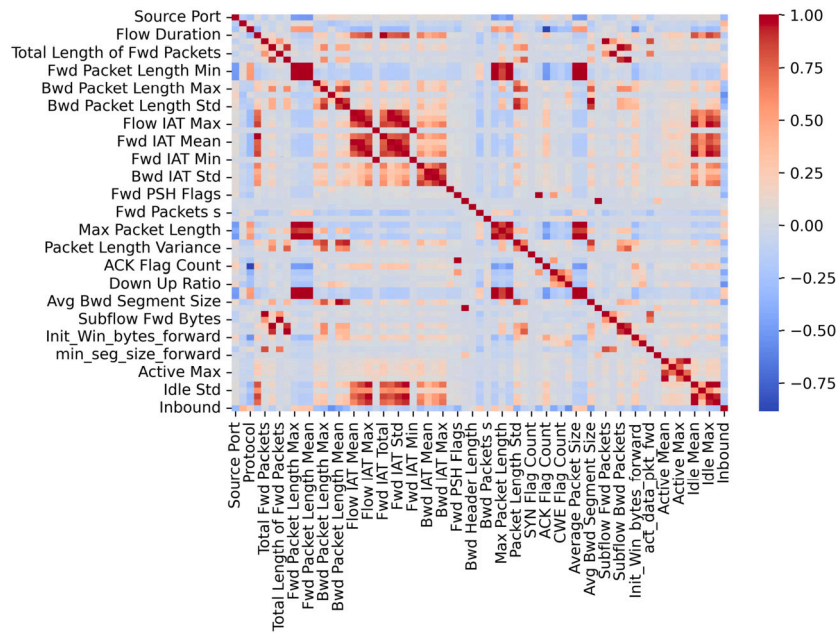


Fig. 6. Correlation analysis of CICDDoS2019 balanced dataset.

analysis was performed to gain a deeper understanding of the behavior of network traffic. This involved gathering data from various network devices, including switches, routers, and firewalls, and using analytical techniques to identify and display irregularities or trends that signal a potential issue. Fig. 6 depicts the data correlation of the preprocessed CICDDoS2019 dataset.

4.2. Performance analysis

To evaluate the effectiveness of our proposed detection and mitigation system, we used various performance metrics, including accuracy, precision, recall, and F1 score. We employed the confusion matrix to calculate the accuracy of each attack type and used the AUC curve for both binary and multi-classification to obtain these metrics.

The accuracy scores and other performance metrics for our system are presented in Figs. 7 to 10. The model demonstrated excellent performance in binary and multi-classification tasks, achieving recall, precision, accuracy, and F1 scores exceeding 99.99% for binary classification and a performance of 98.64% for multi-classification.

In addition to the numerical metrics, we analyzed the system's performance visually using Wireshark and Netdata, an open-source real-time monitoring tool. This allowed us to observe the traffic flow and comprehend the operation of our detection and mitigation system. The I/O flow graph generated by Wireshark (refer to Fig. 13) illustrates the incoming and outgoing traffic patterns, while Netdata provided real-time insights into various key aspects of the system. This included CPU utilization, total IP traffic within the system, anomaly detection, and the aggregate count of IPv4 packets (refer to Figs. 14 to 16). These visualizations enable us to effectively monitor and respond to DDoS attacks by blocking malicious traffic and allowing legitimate traffic to pass through.

4.3. Results

Our proposed work aims to enhance the accuracy and reduce the computational requirements of detecting and classifying DDoS attacks. We first performed data pre-processing before training our model on a multiclass-labeled dataset and evaluated its performance on a separate test dataset using standard metrics.

During the training phase, we optimized our model using the Adam optimizer with a learning rate of $3e-4$ and a batch size of 256. To de-

tect attack packets, we employed the BinaryCrossEntropy loss function, while the CategoricalCrossEntropy loss function was used to classify different types of attacks.

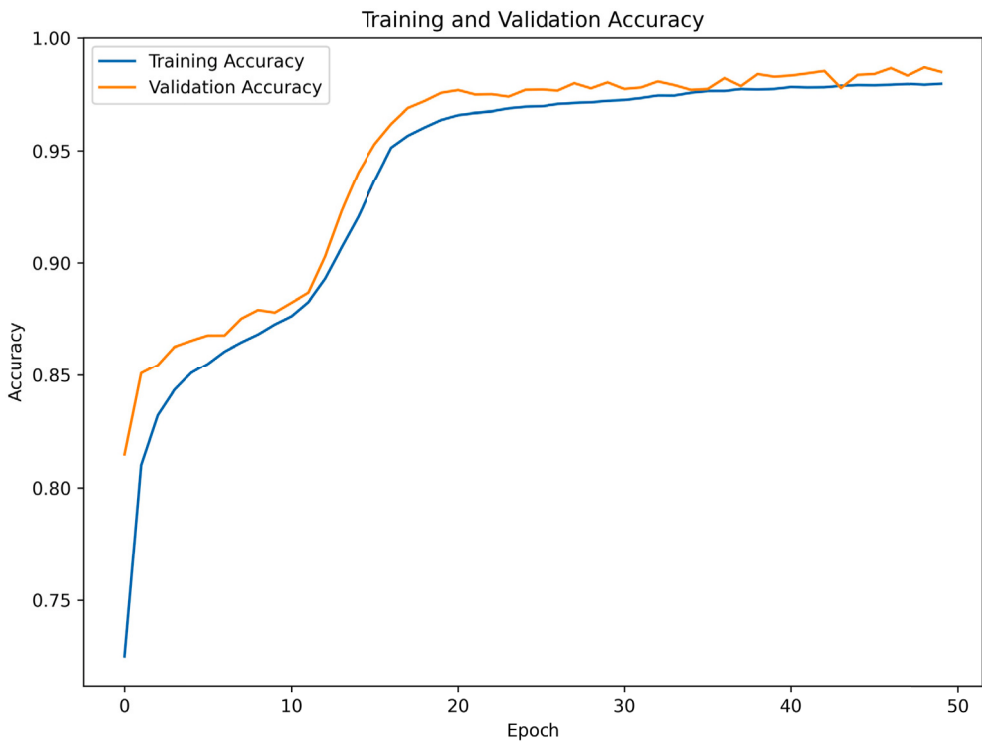
Our model was trained for 50 epochs, with each epoch consisting of 1687 steps due to the large size of the training dataset (431,699 samples). For validation purposes, we split the test dataset into two, one for validation and the other for testing, which contains more than 1.2 lakh records.

The training and validation accuracy, as well as the training and validation losses of the binary and multi-classification models, are depicted in Fig. 7 and Fig. 8.

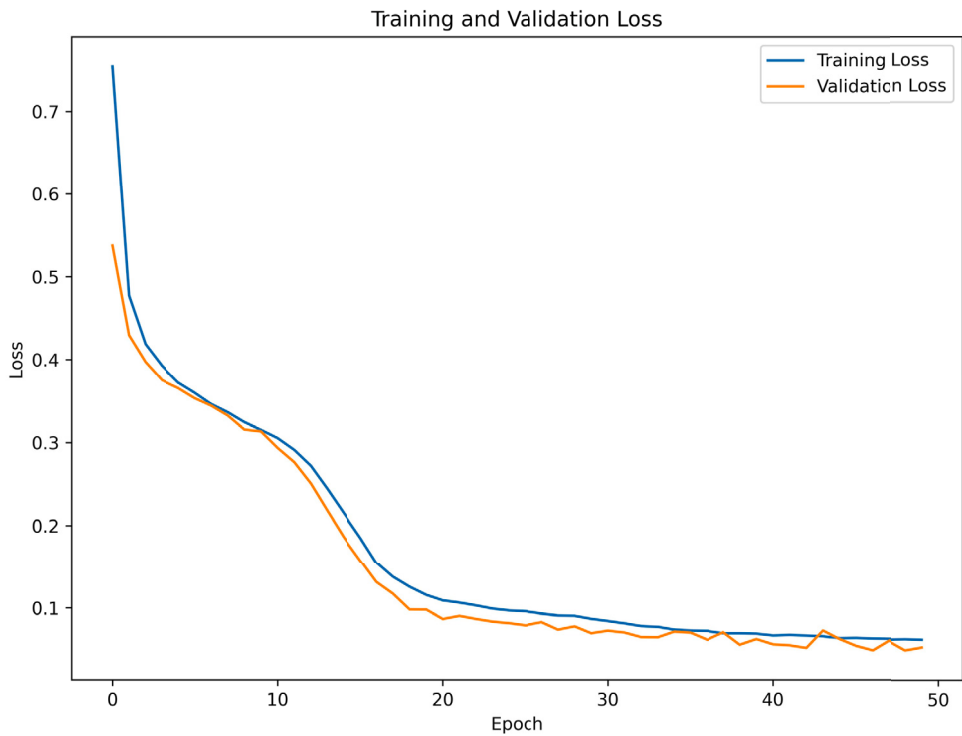
Fig. 7 (a) shows that our model learns exceptionally well, with accuracy increasing rapidly from 70% to 85% within the first 10 epochs and gradually improving to 97.78% after 50 epochs. The validation dataset's accuracy validates our model's effectiveness, achieving an accuracy of approximately 99%. The validation accuracy increased from 82% to 98.05% over 50 epochs, indicating that the model is not overfitting or underfitting but is generalizing well to the dataset. The training and validation loss plots (refer to Fig. 7 (b)) show a decrease in loss from epoch 1 to epoch 50, with a loss of 0.0671 on the training dataset and 0.0496 on the validation dataset after 50 epochs. When tested on a 20% subset of the full dataset, our models achieved an accuracy of 98.64% and precision, recall, and F1 scores of 99%, as shown in Fig. 7.

Fig. 8 (a) shows that our model achieved a training accuracy of 99.76% after 50 epochs on the training dataset, and its accuracy was validated using the validation dataset, achieving an accuracy of approximately 100%. The training and validation loss plots (refer to Fig. 8 (b)) show a decrease in loss from epoch 1 to epoch 50, with a loss of 0.0059 on the training dataset and 0.0017 on the validation dataset after 50 epochs. When tested on a 20% subset of the full dataset, our models achieved an accuracy of 99.99% and precision, recall, and F1 scores of 99.99%, as shown in Fig. 8.

The ROC-AUC is an important evaluation metric for classification models because it provides a comprehensive and threshold-independent measure of their performance. The Receiver Operating Characteristic (ROC) curve is a frequently used performance metric for classification models. When dealing with multi-classification problems, the ROC curve can be constructed by computing the True Positive Rate (TPR) and False Positive Rate (FPR) for each class. In this study, the ROC curve for the multiclass and binary classification problems was calculated, and it produced an overall micro-average AUC of 0.9998% for

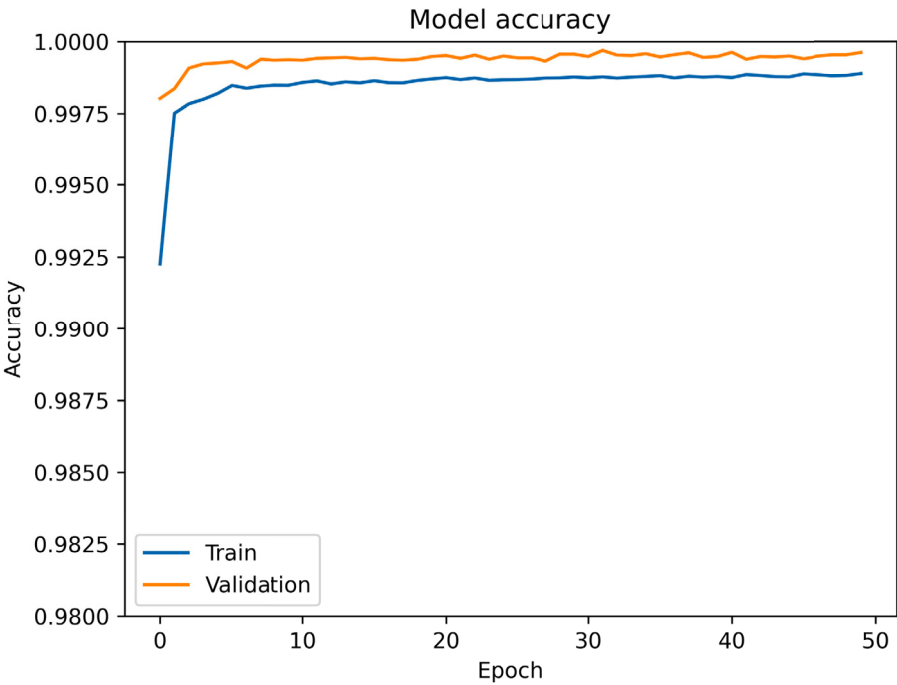


(a) Training and Validation Accuracy

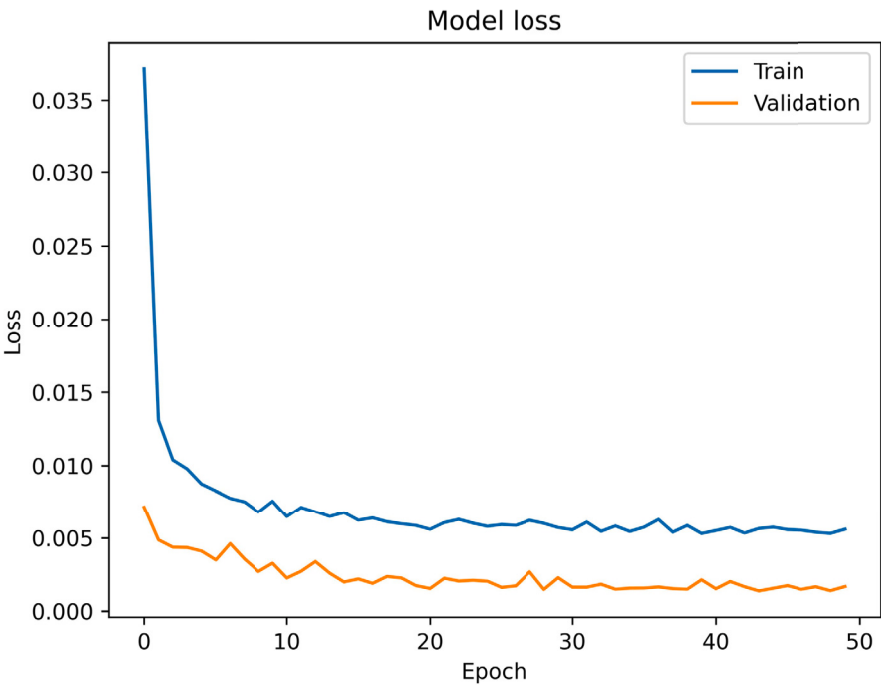


(b) Training and Validation Loss

Fig. 7. Training and validation accuracy and loss for multi-classification.

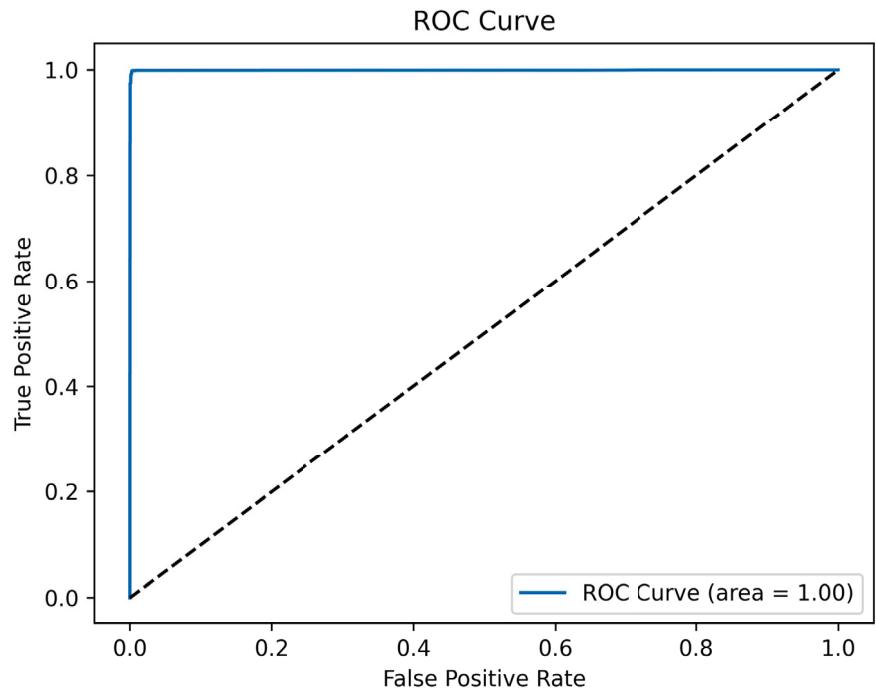


(a) Training and Validation Accuracy

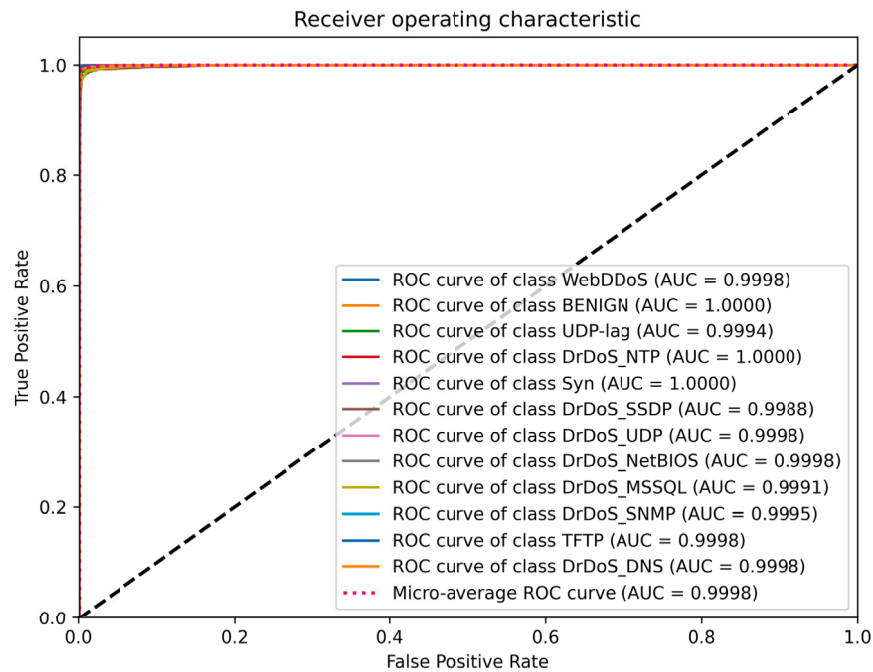


(b) Training and Validation Loss

Fig. 8. Training and validation accuracy and loss for binary classification.



(a) ROC AUC Curve for Binary Classification



(b) ROC AUC Curve for Multi-Classification

Fig. 9. ROC AUC curve for binary and multi-classification.

multi-classification and a 100% score for binary classification. The ROC Curve (AUC) values are depicted in Fig. 9. These AUC values (refer to Fig. 9 (b)) indicate that the classification model performed very well in distinguishing between the different classes, with most of the classes having AUC values close to or equal to 1.0%.

Fig. 10 illustrates the accuracy of various attack types on the CICDDoS2019 dataset for multi-classification. However, classical machine learning and deep learning classifiers have shown lower performance

for class types “WebDDoS” and “SSDP” compared to other types. This is due to the similar characteristics of features between “WebDDoS” and “SSDP,” which implies that additional features are necessary for classifiers to classify traffic records for these classes properly. Despite this, the proposed method achieved an accuracy of 1.00% for “WebDDoS” and 0.97% for “SSDP”. Our model utilizes as many features as possible during training, thus, validating the advantage of the proposed framework through experiments.

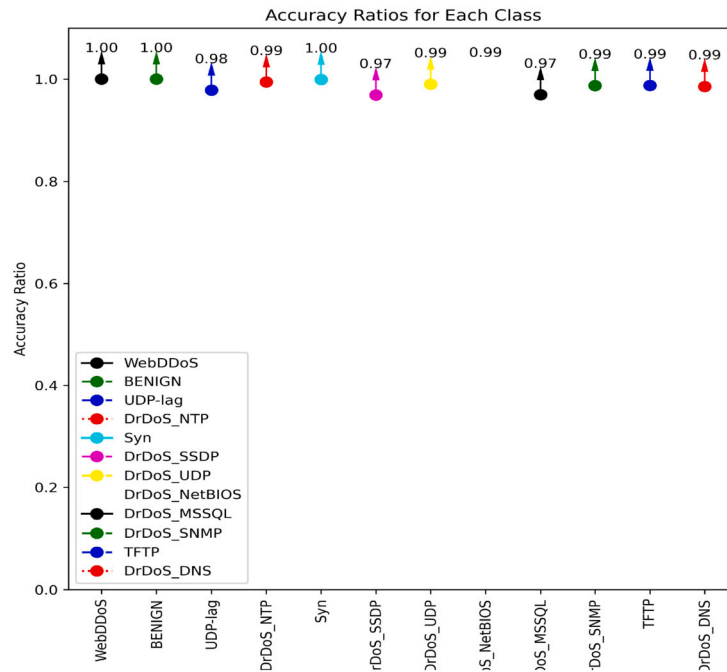


Fig. 10. Accuracy of various attack types on the CICDDoS2019 dataset.

Our proposed model demonstrates high accuracy and efficient computational usage in detecting and classifying DDoS attacks.

5. Simulation (network SetUp and real-time monitoring)

The proposed approach for DDoS detection and mitigation requires a robust testing environment to evaluate its effectiveness in detecting and mitigating attacks while minimizing false positives. However, conducting such experiments in a real-world network environment can be costly and pose significant risks to the network's stability and security.

To address this issue, we utilized Mininet, an open-source Python-based network emulator, and POX controller, an open-source Software-defined networking controller, to create a virtual network environment that simulates various network scenarios. By utilizing Mininet, we were able to create a realistic simulated network topology that includes switches, routers, hosts, and links, allowing us to evaluate the proposed approach in a controlled environment without the need for expensive physical hardware.

The POX controller provided a flexible and extensible platform for programming and managing network traffic in an SDN environment. It allowed us to write our own control applications and algorithms to manage network traffic and supported various network protocols, such as OpenFlow. This made it an ideal choice for our DDoS detection and mitigation experiment.

We designed a custom topology with one controller, 15 switches, and 64 hosts, as shown in Fig. 11. We have trained our CNN model on the CICDDoS2019 Dataset. The model's training encompasses a wide spectrum of DDoS attack types, enabling it to discern subtle variations in network behavior indicative of potential threats. Once trained, the CNN model is deployed on the SDN controller for real-time monitoring. The pre-training process ensures that the model possesses a foundational understanding of various attack patterns, enhancing its ability to swiftly detect and respond to emerging threats as verified on the test dataset, which the model has not seen yet. This integration ensures the model continuously analyses incoming and outgoing traffic, comparing it against the learned patterns. The use of SDN emphasizes the adaptability and programmability of the network, enabling dynamic responses to anomalies without manual intervention. We then installed the proposed DDoS detection and mitigation algorithm in the controller

and generated traffic from regular and attack sources to evaluate the approach's efficacy. To simulate a DDoS attack, we configured one of the Hosts (h1) to generate traffic with spoofed IP addresses using the hping3 tool, while hosts h16 and h31 generated normal traffic.

Periodically, the POX controller retrieved flow entry information through `flow_stats_request` and `tcpdump` to detect and mitigate the DDoS attack effectively. When a DDoS attack is detected, a detailed notification email is sent to the administrator's registered email address. The algorithm then analyzed the network traffic for IP spoofing. If a spoofed IP address was found, the algorithm (Algorithm 3) ran a subprocess command to add a blocking rule in the INPUT chain of iptables, instructing it to drop any incoming packets from the detected spoofed IP address. This effectively neutralized the threat posed by the spoofed IP. However, if a spoofed IP address was not found, the mitigation system applied Algorithm 2. This algorithm effectively countered the DDoS attack by inserting a block flow rule on the switches, preventing the malicious traffic from congesting the network. The algorithm employed a rate limit mechanism to optimize the IP blocking process and avoid unnecessary disruptions to legitimate traffic. This mechanism limits the number of IP addresses that can be blocked in a given period of time. This helps to ensure that only malicious IP addresses are blocked while legitimate traffic is not affected (Fig. 4). The experiment results are depicted in Fig. 12, and a description of all the parameters used in the experiment is listed in the Table 4.

Overall, the simulation setup using Mininet and POX controller was critical in providing a realistic testing environment for evaluating the proposed DDoS detection and mitigation approach. By creating a virtual network environment, we could test the approach without incurring significant costs associated with physical hardware and minimize the risk of instability and security breaches in a real-world network environment. The email alert feature ensures that network administrators are promptly notified of DDoS attacks, which can help them understand the attack and enhance the system for future challenges. The results of the experiment demonstrated the effectiveness of the proposed approach and its potential to improve network security in the face of DDoS attacks.

Fig. 13 is an I/O graph generated through Wireshark, a widely used network protocol analyzer. The graph visually represents the network traffic flow during a DDoS attack, with the x-axis indicating the time

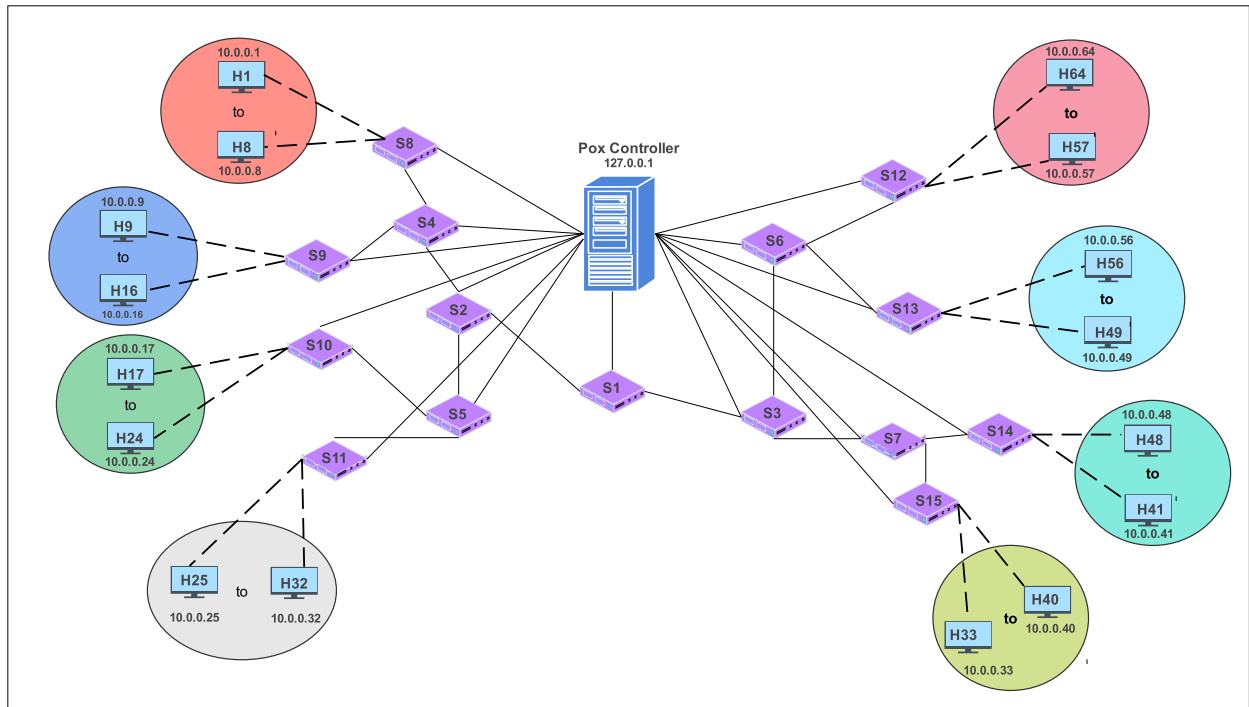


Fig. 11. SDN topology.

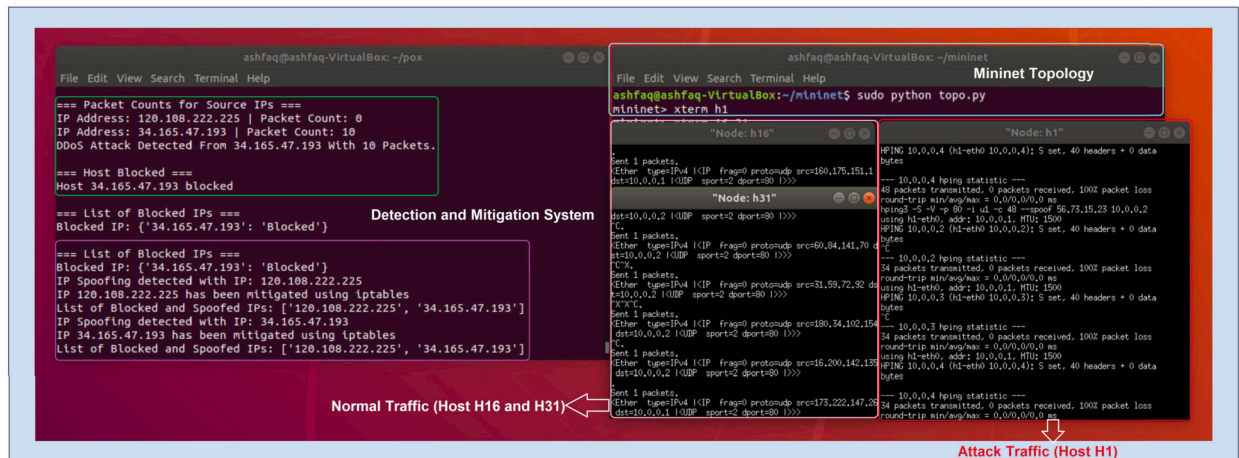


Fig. 12. Results of simulation.

and the y-axis indicating the packets/sec. The graph in Fig. 13 clearly shows the impact of the DDoS attack on the network's traffic flow.

The attack started attacking h2 at 04:17:11 IST, which is evident from the initial increase in the graph. However, the quantity of h2 packets started decreasing soon after, and the traffic returned to its normal state, indicating the attack's negative impact on the network's performance and the effectiveness of our mitigation action in restoring the network's normal traffic flow.

We conducted a series of experiments to assess the effectiveness of the proposed model across three distinct scenarios: Attack-Free, Attack-No Mitigation, and Attack-Mitigation. The results for these experiments were extracted using Netdata, a real-time monitoring tool. The outcomes of these experiments are depicted in Fig. 14, Fig. 15 and Fig. 16.

In Fig. 14 (a), we illustrate our experimental assessment of CPU utilization across three distinct phases. In the initial phase, we generated regular traffic using the Scapy tool and observed stable CPU utilization, ranging from 38% to 40%. Transitioning to the second phase, we gener-

ated attack traffic with spoofed source IPs, resulting in a rapid increase in CPU utilization to 60%, further intensifying to 62.7%. This escalation was attributed to the disruptive nature of the attack traffic, which led to reduced predictability and less efficient use of CPU resources.

Subsequently, we implemented the proposed mitigation system and replicated the experiment. In the attack scenario with the mitigation system, CPU utilization witnessed a significant reduction. It initially dropped to 33.8%, followed by further declines to 31.9% and ultimately reaching 6.9% over time. These results demonstrate the mitigation system's effectiveness in curtailing CPU utilization during a DDoS attack.

Importantly, the absence of the proposed mitigation system could give rise to several issues. These include the potential partial failure of the controller, overload in OpenFlow switch tables and controller's flow rules, dropped malicious requests, disconnection of the attacker's associated OpenFlow switch, and potential system destabilization. These graphs offer valuable insights into the impact of DDoS attacks on sys-

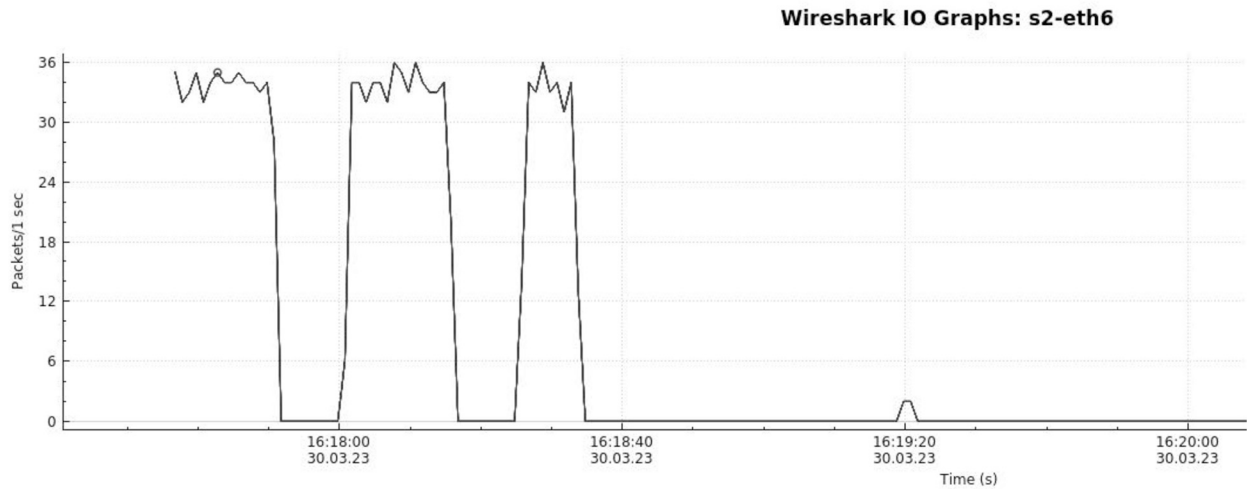


Fig. 13. IO flow graph.

Table 4
Parameters used in the simulation.

Sl. No	Parameters
1	Host: Windows 11
2	Guest: Ubuntu 18.04
3	VM Virtual Box: 7.0
4	Emulator: Mininet
5	Controller: Pox
6	Number of Controller: 1
7	Number of Switches: 15
8	Number of Hosts: 64
9	Protocol: OpenFlow
10	Visualization: MiniEdit
11	Port for Controller: 6633
12	IP address for Host H1: 10.0.0.1
13	IP address for Host H64: 10.0.0.64
14	Controller-to-switch protocol: OpenFlow 1.3
15	Bandwidth: 100 Mbps
16	Delay: 10 Ms
17	Queue size: 1000 packets
18	Packet loss: 0%
19	Traffic type: Regular and DDoS
20	DDoS attack type: IP spoofing
21	DDoS attack tool: hping3

tem performance and the effectiveness of the proposed detection and mitigation system in addressing such attacks.

Fig. 14 (b) visualizes the flow of total IP traffic in the system under different scenarios. In the Attack-Free scenario, the system receives 254 Kilobits Per Second (kbps) of traffic and sends -254 kbps of traffic. In the Attack - No Mitigation scenario, the amount of incoming traffic spikes to 1.31 Megabits Per Second (Mbps), and the amount of outgoing traffic spikes to -1.31 Mbps. This is because the attack traffic is not being mitigated, so it overwhelms the system.

The Attack - No Mitigation scenario could have serious consequences. The high traffic volume could strain the system's resources and cause performance degradation or service disruption. The system may not be able to process incoming packets efficiently, which could lead to delays or data loss. This scenario shows how important it is to have effective mitigation measures to protect the system from DDoS attacks.

In contrast, the Attack- Mitigation scenario shows how effective mitigation can be. The system can manage the incoming traffic and keep it manageable, even in the face of a DDoS attack. The incoming traffic is reduced to 731 kbps, and the outgoing traffic is reduced to -731 kbps. This ensures that the system is not overwhelmed and can continue operating normally. The mitigation approach also stabilizes total IP traffic. It protects the integrity of SDN, keeping the amount of traffic coming into and going out of the system at a constant level.

The efficacy of our proposed DDoS detection and mitigation system can be assessed by looking at the anomaly detection metrics presented in Fig. 15. The anomaly rate is a pivotal measure of the detection and mitigation system's accuracy. A higher anomaly rate indicates increased sensitivity to detecting deviations from normal behavior, potentially resulting in false positives. These metrics show how well the system can distinguish between normal and anomalous behavior.

In the Attack-Free scenario, our proposed system identifies 1077 dimensions as normal and 0 as anomalous. This results in an anomaly rate of 0%, which is very good. This shows that our system is very accurate in distinguishing between benign and anomalous traffic patterns.

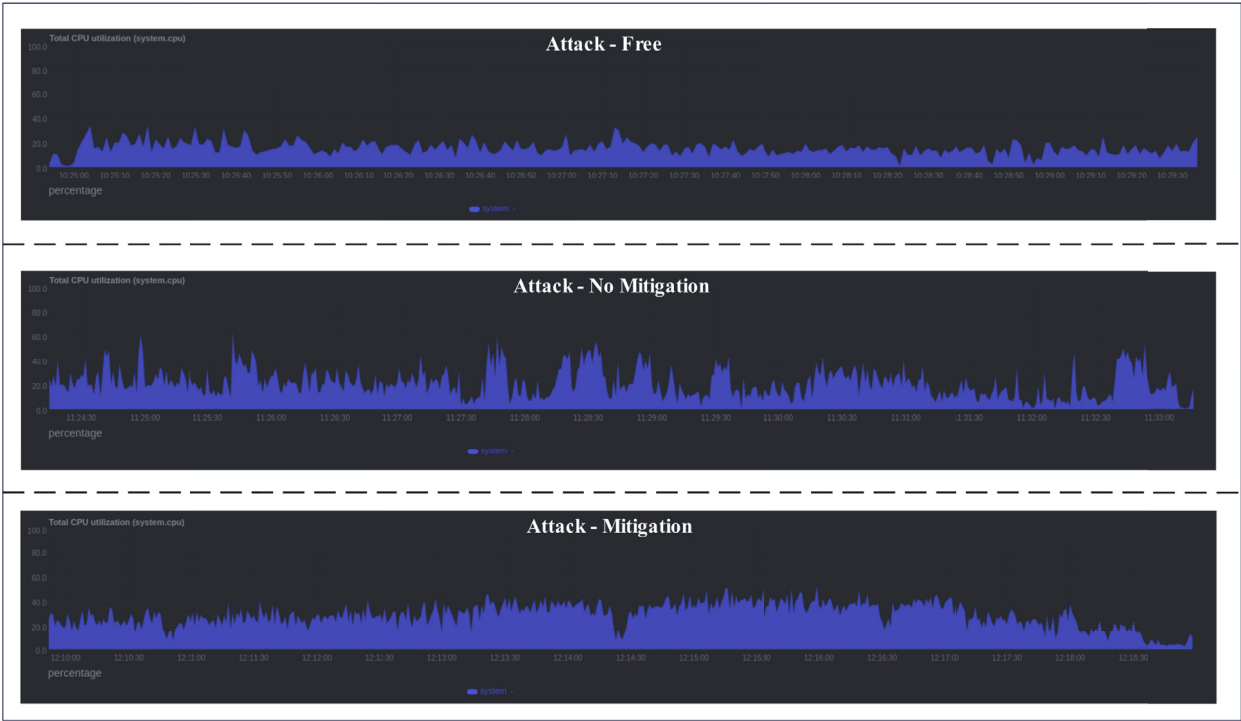
In the Attack-No Mitigation scenario, our proposed system identifies 33 as anomalous out of 3170 dimensions. This translates to an anomaly rate of 0.9%. This indicates that our system starts detecting deviations in behavior at a relatively modest rate.

The Attack-Mitigation scenario shows the best results. At 12:13:09 IST, our proposed system detects 3163 dimensions as normal and identifies 41 as anomalous. The resulting anomaly rate of 1.37% signifies the system's efficacy in detecting anomalous behavior while minimizing false positives. This precision is further exemplified at 12:16:40 IST, with 3202 normal dimensions and a mere two anomalous dimensions, yielding an anomaly rate of 0.28%. Most impressively, effective mitigation attains perfection in anomaly detection. At 12:18:00 IST, our proposed system achieves a flawless 0% anomaly rate. This shows that the system is capable of flawlessly discerning between normal and anomalous dimensions.

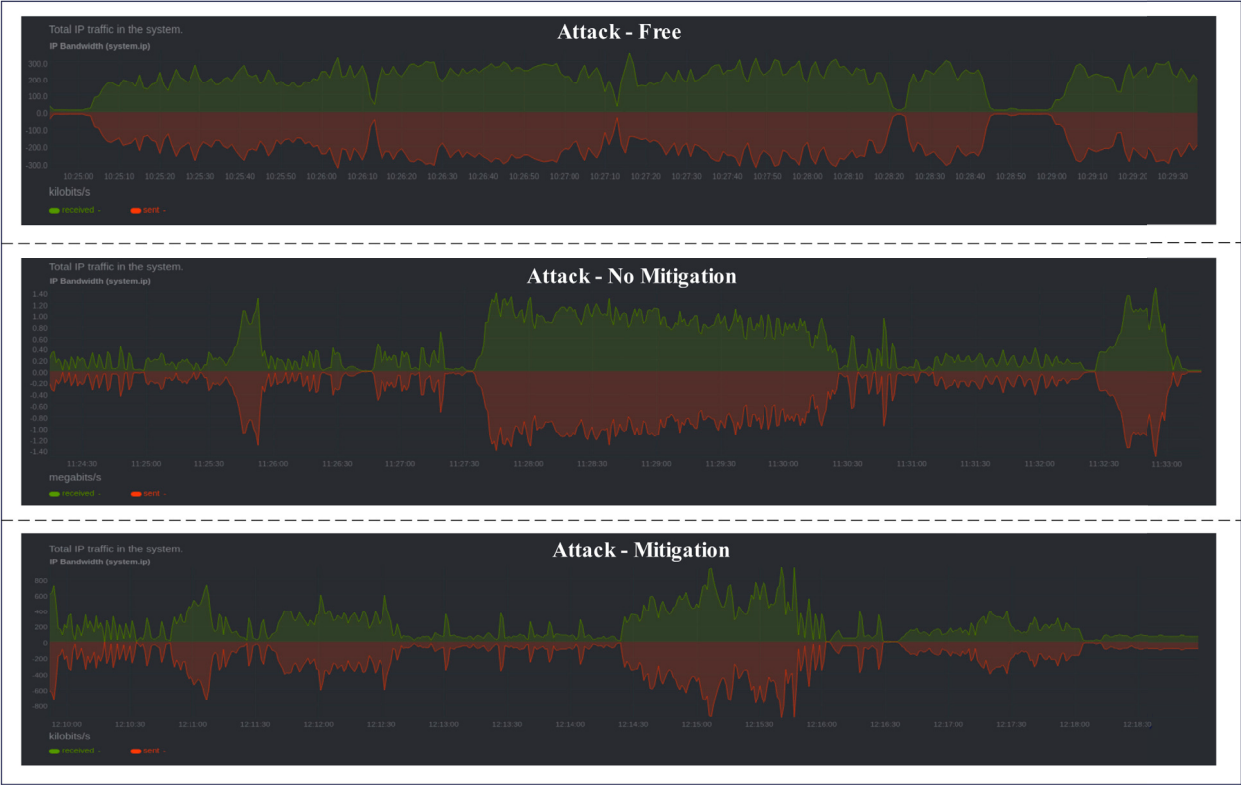
Our proposed DDoS detection and mitigation system has a higher detection rate than traditional approaches. In the attack mitigation scenario, our system achieves a 0% anomaly rate, which is significantly better than the traditional approaches. This shows that our system is more effective at detecting DDoS attacks and minimizing disruptions to the network.

Fig. 16 illustrates the changes in IPv4 packet metrics across different scenarios. In the Attack-Free scenario, the network operates normally with an average packet reception rate of 50.7 pps. The peak reception rate is 63.6 pps, and the sending and delivery rates are the same. In the Attack-No Mitigation scenario, the network is overwhelmed by a DDoS attack, and the packet reception rate spikes to 1.65 kpps. The sending and delivery rates drop significantly, which could lead to network congestion, delays, and even crashes.

In the Attack-Mitigation scenario, the mitigation strategies can stabilize the packet rate. The average packet reception rate is 1.02 kpps, and the sending and delivery rates are much closer to the reception rate. This ensures that the network can continue to operate normally, even during a DDoS attack.

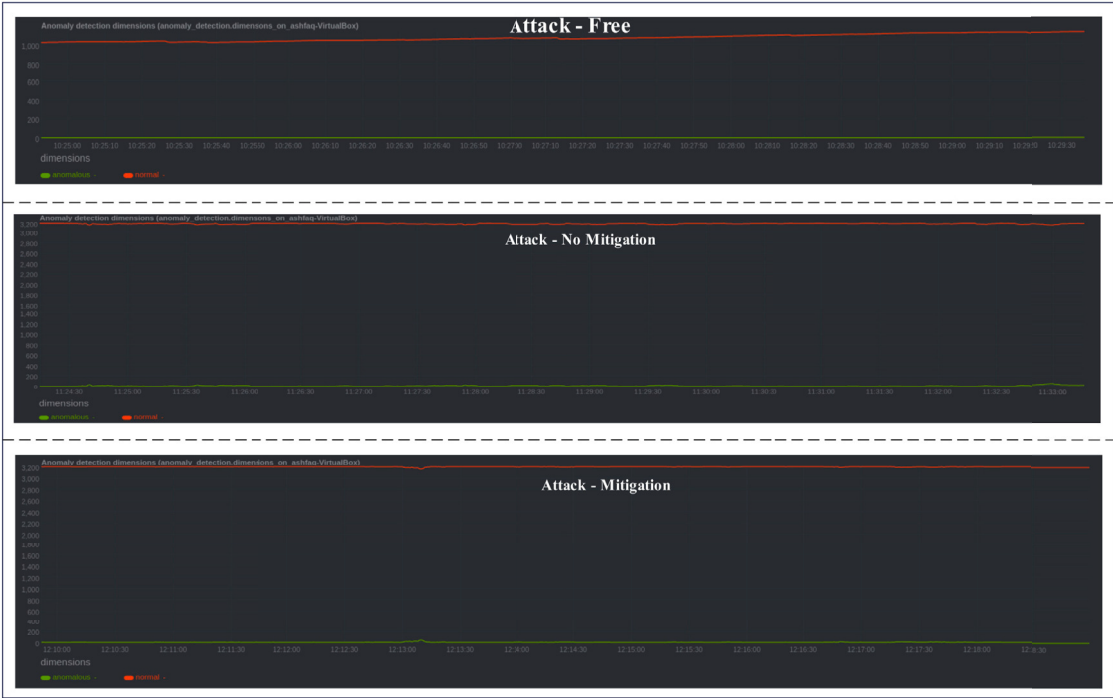


(a) CPU Performance during Network Scenarios

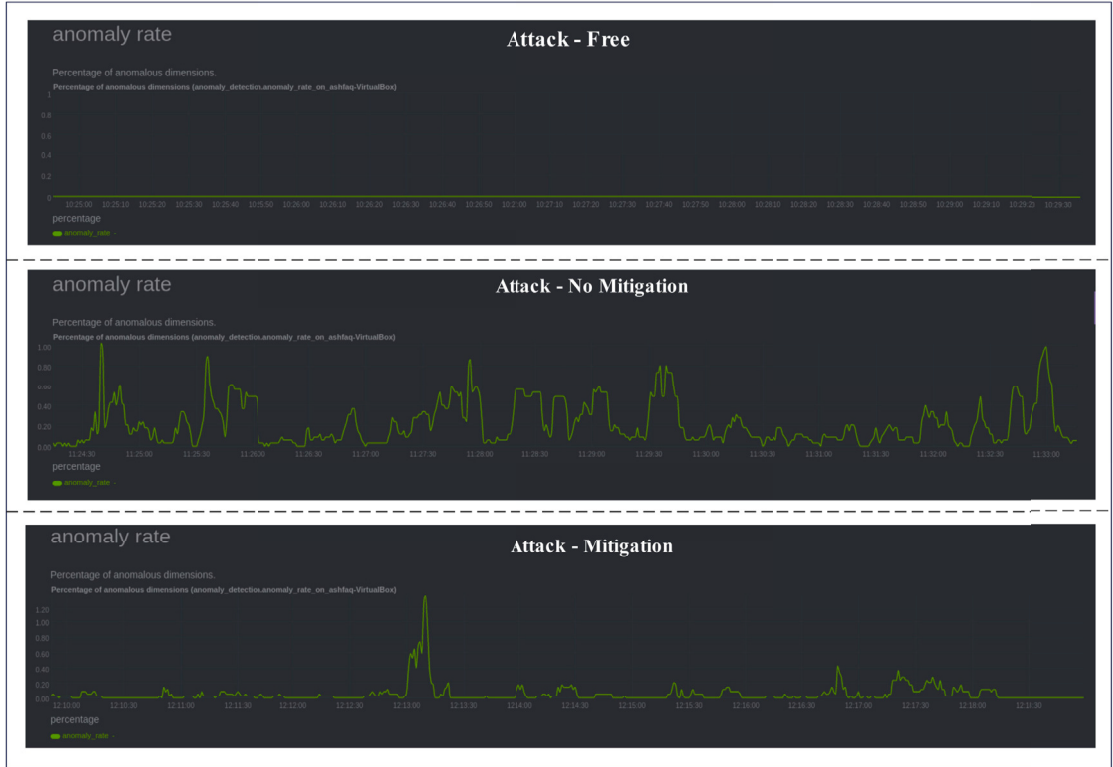


(b) Total IP Traffic In a System

Fig. 14. CPU performance and total IP traffic.



(a) Total Dimensions Normal and Anomalous



(b) Anomaly rate

Fig. 15. Anomaly detection based on dimensions and anomaly rate.

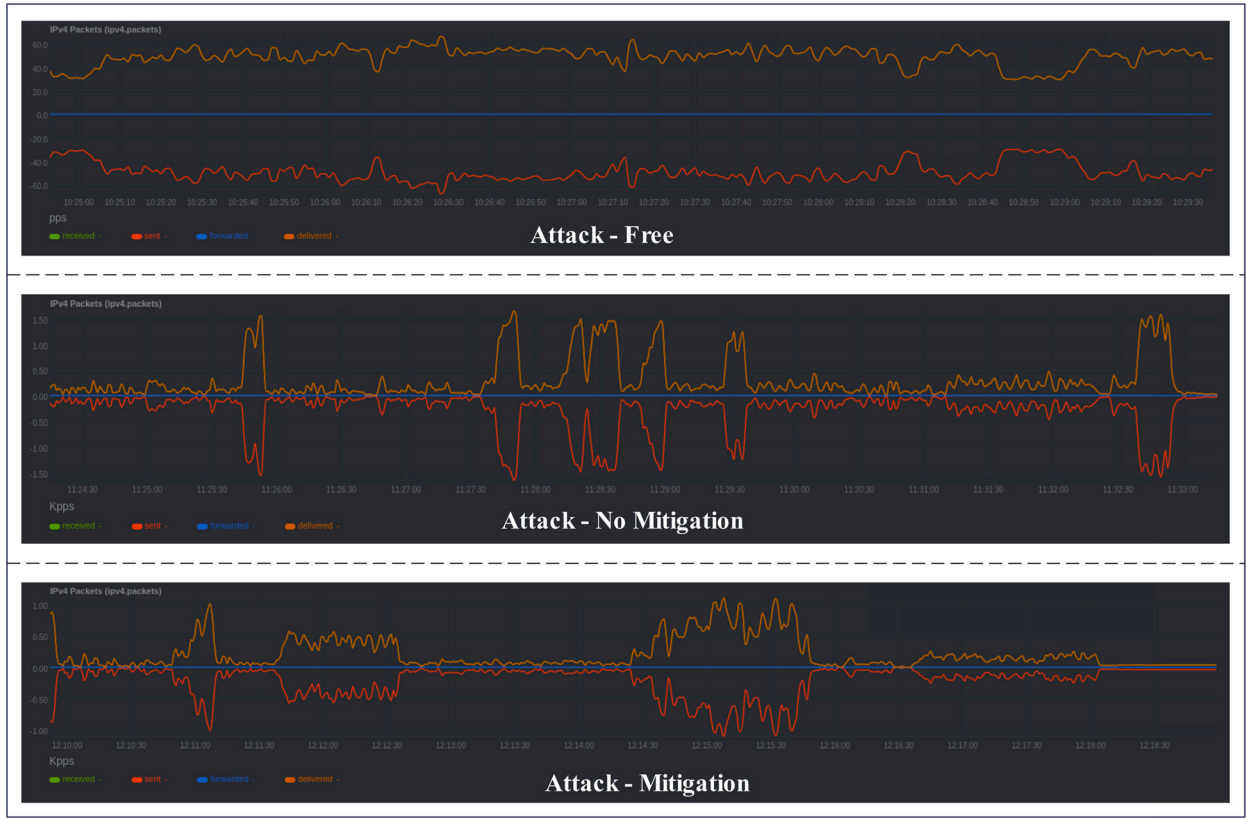


Fig. 16. Total IPv4 packets in a system.

Furthermore, we conducted a series of experiments to evaluate the effectiveness of our proposed DDoS detection and mitigation system. The results are depicted in Fig. 17. In the first scenario, we examined the system's performance without the mitigation module to observe its effects. As depicted in Fig. 11, our network topology comprises 15 switches, 64 hosts, and one controller. In this experiment, h1 served as the attack initiator, while h2 represented the victim, and h16 and h31 sent normal traffic requests to h2.

To detect DDoS and normal attack traffic, we implemented our DDoS detection algorithm, which was discussed in (subsection 3.2). Upon identifying an attack, the user received an email notification to their registered email ID. Meanwhile, normal flow table operations continued undisturbed.

Fig. 17 (a) illustrates the results when we used hping3 to launch a flooding attack on h2, using h1. Consequently, h16 and h31 could not send data requests to h2, and the controller promptly alerted the user about the ongoing attack on h2.

We activated the mitigation strategies in the second test scenario. We configured the traffic rule to drop packets corresponding to a specific input port to mitigate the flood attack effectively. Consequently, in Fig. 17 (b), we observed that h31 and h16 could send data to h2 normally after mitigation.

These results demonstrate the effectiveness of our proposed mitigation system in countering DDoS attacks and ensuring the continuity of normal network operations.

6. Comparison with existing literature

Our proposed deep-learning model for detecting and classifying DDoS attacks in SDN was compared with existing models, as shown in Table 5. Our model outperformed others with significantly higher accuracy, precision, recall, and F1 score. For binary classification, our model achieved 99.99% precision, recall, and F1 score, along with 100% AUC

ROC. For multi-classification, our model achieved 98.64% accuracy, 99% precision, recall, and F1 score, and 99.97% AUC ROC.

Moreover, Fig. 18 compares the proposed model with the baseline model to evaluate the effectiveness of our proposed model. It is visually depicted in the figure that the proposed model has surpassed all other models in terms of performance.

Our proposed model demonstrates exceptional performance in binary and multi-classification scenarios. It is computationally efficient, with a lower number of parameters compared to other models. These results highlight the effectiveness and practicality of our model for detecting and classifying DDoS attacks in SDN environments.

6.1. Computational cost analysis in comparison with existing models

The consideration of computational cost holds significant importance in DDoS detection, directly influencing both the efficiency of the detection system and the resources needed for its operation. As the frequency and intricacy of DDoS attacks continue to rise, the real-time detection process becomes resource-intensive. The computational cost of the detection system can emerge as a bottleneck, affecting not only the system's efficacy but also the overall performance of the underlying network. A detection system with excessive computational demands may struggle to cope with the traffic demands of a high-volume DDoS attack, potentially resulting in service disruptions or downtime. Conversely, a simplistic detection system may fail to differentiate between legitimate and malicious traffic, leading to false positives and unnecessary mitigation actions. In the context of Software Defined Networks, where network configurations and policies are dynamically controlled through software, a computationally excessive detection system can impede the agility and responsiveness of the SDN infrastructure. SDNs rely on efficient and rapid decision-making to dynamically adapt to changing network conditions. If the detection system consumes too many computational resources, it may hinder the SDN's ability to quickly

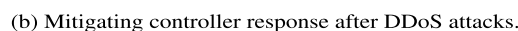
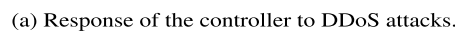


Fig. 17. Impact of the controller to DDoS attack.

Model Comparisons

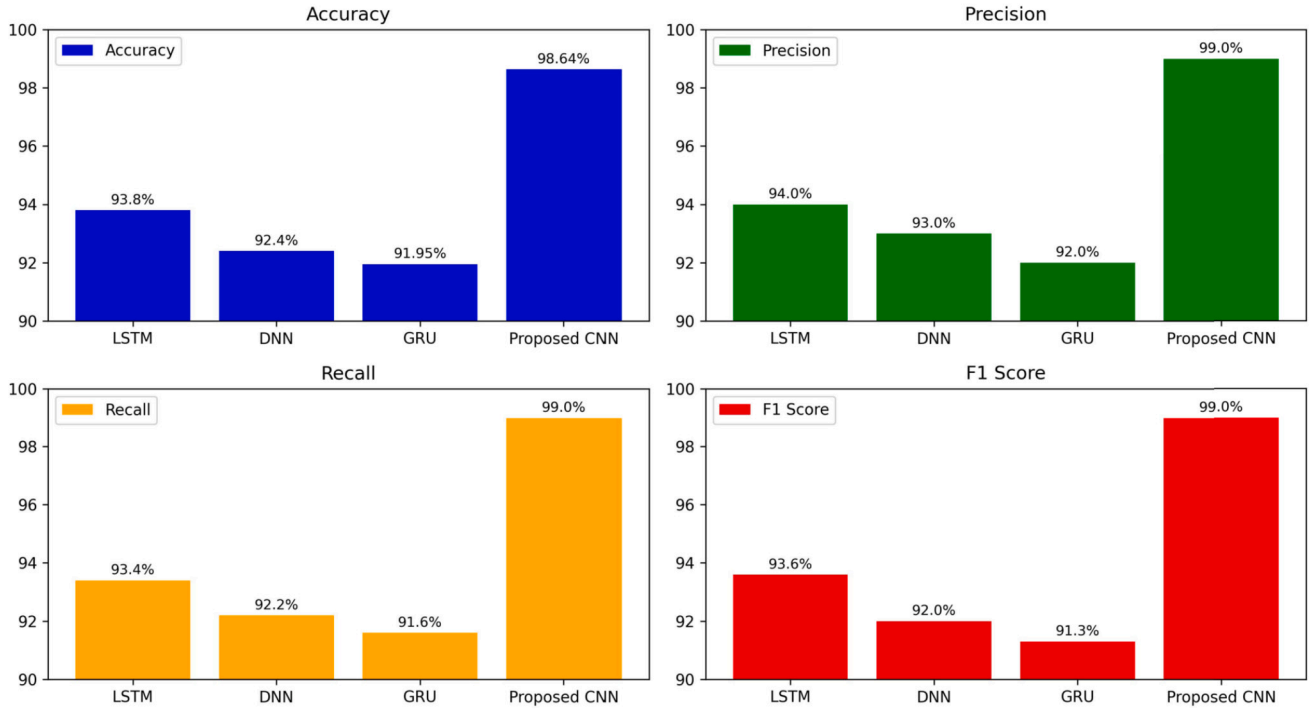


Fig. 18. Comparison of different baseline models with proposed multi-classification model.

Table 5

Comparison of various DL approaches on CICDDoS2019 dataset.

Reference	Model	Dataset Used	Results	Detection	Mitigation
Jagtap and Saravanan (2022)	LSTM-GRU & SVM	CICDDoS2019	Accuracy of 98.5% (B) & 95.5% (M)	Yes	No
Elsayed et al. (2020b)	RNN + Autoencoder	CICDDoS2019	F1 score 99%	Yes	No
Cil et al. (2021)	DNN	CICDDoS2019	Accuracy of 99.5%(B) & 94.5%(M)	Yes	No
Ferrag et al. (2021)	CNN	CICDDoS2019	Accuracy of 95.12% (M)	Yes	No
Novaes et al. (2020a)	LSM-FUZZY	CICDDoS2019	Precision of 97.89%(B)	Yes	Yes
Yungaicela-Naula et al. (2021)	CNN	CICDDoS2019	Detection rate of 95% on Slow rate& 98% on High volume attacks(B)	Yes	No
de Assis et al. (2020)	CNN	CICDDoS2019	Accuracy of 95.4 (M)%	Yes	Yes
de Assis et al. (2020)	MLP	CICDDoS2019	Accuracy of 92.5(M)%	Yes	Yes
Proposed Model	BRS + CNN	CICDDoS2019	99.99%(B) & 98.64%(M)	Yes	Yes

B: Binary Classification, M: Multiclassification.

implement countermeasures or route traffic optimally in response to a DDoS attack.

Table 6 presents a comprehensive analysis comparing various models utilized for DDoS attack detection, including the model proposed in this study. Each entry in the table delineates the reference, model architecture, dataset used, evaluation metrics (accuracy, loss, and number of classes), as well as time-related metrics (training and inference durations). In contrast to the referenced studies, the 1D-CNN model proposed in this study demonstrates competitive or superior accuracy across all datasets. Notably, the training and inference times are significantly reduced, underscoring the model's efficiency. The 1D-CNN architecture employed in this study achieves comparable or better results with a decreased computational overhead, making it a promising approach for real-time DDoS attack detection in SDN environments. Our proposed approach not only tackles computational cost concerns but also excels in both binary and multi-classification scenarios, showcasing superior performance.

The effectiveness of the CNN model in detecting DDoS attacks has been empirically validated, affirming its practical applicability. To advance this capability, the next crucial step involves implementing DDoS attack detection within the SDN paradigm. In pursuit of this goal, our chosen approach entails the utilization of a pre-trained CNN model. This model has been trained using the CICDDoS2019 dataset, a comprehensive and representative dataset encompassing new types of DDoS attacks. The integration of the CNN model into the SDN framework is orchestrated through the PoX controller, serving as a pivotal component. The PoX controller facilitates seamless communication and coordination between the CNN model and the SDN infrastructure. This strategic amalgamation aims to leverage the discriminative capabilities of the CNN model to discern normal network behavior from anomalous patterns indicative of potential DDoS attacks. The use of a pre-trained model enhances the efficiency and accuracy of DDoS detection within the SDN ecosystem, providing a sophisticated layer of defense against evolving cyber threats.

Table 6
Computational cost analysis in comparison with existing models.

Reference	Model	Dataset	Evaluation Metrics		Time (in seconds)		Accuracy
			No of Classes	Loss	Training	Inference	
Elmasry et al. (2020)	PSO + DNN	CICIDS2017 (B)	2	0.28	1274 s	637 s	97.85%
	PSO + LSTM-RNN			0.16	1492	846	98.83%
	PSO + DBN			0.1	1617 s	915 s	99.91%
Elmasry et al. (2020)	PSO + DNN	CICIDS2017 (M)	8	0.28	1911 s	956 s	88.04%
	PSO + LSTM-RNN			0.16	2238 s	1269 s	98.41%
	PSO + DBN			0.1	2426 s	1373 s	95.81
Marvi et al. (2021)	LGBM/PM-Model	CICDDoS2019	SC	*	<3 mins	*	99.96%
	LGBM/LDAP-Model		SC	*	<5 mins	*	99.9%
	LGBM/SYN Model		SC	*	<10 mins	*	99.98%
Dong et al. (2021)	SSDDQN	NSL-KDD (B)	2	*	398 s	*	83.01%
		AWID (B)	2	*	*	*	98.99%
		NSL-KDD (M)	5	*	*	*	79.43%
		AWID (M)	4	*	*	*	98.19%
Mauro et al. (2020)	WiSARD	CIC-IDS-2017/2018	SC (DDoS)	*	104.783 s	*	98.80%
	WiSARD	CIC-IDS-2017/2018 (M)	SC (PortScan)	*	106.68 s	*	99%
			4	*	119.848 s	*	89%
This Study	BRS + CNN	CICDDoS2019 (B)	2	0.0017	156.40 s	0.2633 s	99.99%
		CICDDoS2019 (M)	12	0.0496	194.95 s	0.3697 s	98.64%

* Not available, SC: Single-Class, B: Binary Classification, M: Multi Classification.

7. Conclusion and future work

DDoS attacks pose a significant threat to online services and applications, making it crucial for organizations to strengthen their defense mechanisms continually. In this paper, we propose a novel approach (BRS + CNN) for DDoS attack detection in SDN environments. We developed a balanced dataset using a random sampling technique to address the data imbalance problem commonly encountered in the field. This approach helped to balance the distribution of data across classes and improve the performance of our model. From the experimental results, the proposed model achieves high performance in binary and multi-classification tasks. For binary classification, the model achieves an accuracy of over 99.99% and an F1 score of 0.9999%. For multi-classification, the model achieves an accuracy of 98.64% with an F1 score of 0.99%. To mitigate these threats, we have implemented a variety of mitigation techniques, including filtering, rate limiting, and the utilization of iptables rules for the purpose of blocking spoofed IP addresses. We also introduced a monitoring system that utilizes rate-limiting to ensure that legitimate traffic is processed efficiently. Our proposed DDoS detection system not only detects the attack but also sends detailed contextual information to a designated email address. This information will help administrators to know more about the attack and can enhance the system for future challenges. We evaluated the efficiency and effectiveness of our proposed DDoS mitigation system by conducting a series of experiments across three distinct scenarios. We tested its effectiveness through a real-time monitoring tool (Netdata) and found it reliable in safeguarding networks against DDoS attacks while maintaining regular operations. The results demonstrated the robustness of our proposed mitigation system in effectively combating DDoS attacks while also safeguarding the seamless continuity of regular network operations.

Future research should focus on improving the scalability and flexibility of DNN-based detection systems for large-scale SDN networks. Distributed computing and parallel processing techniques could be explored to speed up model training and evaluation. Furthermore, it is essential to evaluate the effectiveness of DNN-based detection systems in various SDN environments and network topologies to develop more efficient and targeted defense mechanisms against DDoS attacks. By continuing to stay updated on the latest developments in DDoS attack detection, organizations can better protect themselves and their customers from the harmful consequences of such attacks.

CRedit authorship contribution statement

Ashfaq Ahmad Najar: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing.
S. Manohar Naik: Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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